

RESEARCH ARTICLE

DBC-LSTM-ARIMA as a Dynamic Bi-Directional Coupling and Jointly-Optimized Hybrid Approach for Stock Price Prediction

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ABSTRACT Accurate stock price prediction is urgently needed for economic and algorithmic investment, yet prices are highly non-linear, non-stationary, and regime-switching, which makes reliable forecasting very challenging. To address this issue, we propose Dynamic Bi-directionally Coupled Long Short-Term Memory–AutoRegressive Integrated Moving Average (DBC-LSTM-ARIMA), a hybrid model that couples an AutoRegressive Integrated Moving Average (ARIMA) backbone with a Long Short-Term Memory (LSTM) network under a non-anchored Walk-Forward Optimization (WFO) protocol. In each sliding window, ARIMA first models short-run linear dynamics and produces one-step forecasts and residuals. These residuals, together with price history and market covariates, are fed forward into the LSTM, while the LSTM’s one-step prediction errors are injected back into ARIMA as an exogenous correction, forming a bi-directional feedback loop. Both components are trained with a unified window-level loss and alternating updates, and their outputs are combined through an adaptive convex fusion weight determined from recent errors. Extensive experiments on three Chinese A-share stocks (BYD, China Unicom, Ping An Bank), three overseas blue-chip stocks (BP p.l.c., Apple Inc., Citigroup Inc.), and six additional A-share stocks from diverse industries show that DBC-LSTM-ARIMA consistently outperforms seven state-of-the-art baselines in terms of Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), Mean Absolute Percentage Error (MAPE), and coefficient of determination (R^2), while maintaining moderate computational cost suitable for rolling deployment.

INDEX TERMS Stock price prediction, ARIMA-LSTM hybrid, bi-directional coupling, joint optimization, walk-forward evaluation.

I. INTRODUCTION

With financial globalization accelerating, equity markets have become more competitive, higher-frequency, and structurally complex [1] (O’Hara, 2015). In this environment, reliable stock price prediction is central to asset allocation, risk control, and policy guidance, because more accurate forecasts translate into fewer adverse trades, more stable portfolio rebalancing, and more disciplined exposure management [2]. At the same time, individual stock prices

typically display strong non-stationarity, frequent regime shifts, and substantial noise contamination, which together undermine the effectiveness of standard forecasting pipelines and complicate the extraction of exploitable signals from historical data [3]. As a result, designing prediction frameworks that can remain robust under changing market conditions while capturing both short-term dynamics and longer-horizon structures has become a core challenge in modern quantitative finance.

Classical econometric models—Autoregressive (AR), Autoregressive Moving Average / Autoregressive Integrated Moving Average (ARMA/ARIMA), and Generalized

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Autoregressive Conditional Heteroskedasticity (GARCH)—remain strong at capturing short-horizon linear dynamics and volatility clustering [4], [5], [6]. They require careful differencing, stationarity checks, and order selection, and they struggle when market dynamics become non-linear or when structural breaks dominate the signal. As a result, pure ARIMA or GARCH pipelines often suffer from lagged responses and systematic bias in trending or regime-switching episodes [7], [8], [9].

Machine learning and deep learning (e.g., Random Forests, SVMs, RNN/GRU/LSTM) improve non-linear feature extraction and long-memory modeling, but they are susceptible to overfitting and unstable signals under volatility spikes [10], [11], [12]. Hybrids that feed ARIMA residuals into an LSTM have been proposed to combine linear and non-linear structure while using walk-forward evaluation; in these designs the final forecast is produced solely by the LSTM and information flows only from ARIMA to LSTM, without any feedback into ARIMA or joint optimization of both parts. A recent representative is the LSTM-ARIMA strategy [13] that appends ARIMA residuals and market covariates to the LSTM input, evaluates models in a walk-forward process across major indices, and reports risk-adjusted metrics such as modified information ratio.

Despite the progress, this line of research still belongs to a broader class of methods that either employ ARIMA alone, LSTM alone, or a one-way ARIMA-LSTM hybrid. There are still some challenges to be addressed:

- (i) Residual information flows in only one direction, leaving ARIMA without any feedback correction and causing error accumulation during turning points or volatility spikes;
- (ii) ARIMA and LSTM are trained under separate objectives, leading to parameter drift and potential conflicts in trading signals;
- (iii) Forecasts are produced by a single model output or by static weighting, without adapting the combination to changing market regimes.

Addressing these shortcomings is essential for improving predictive accuracy, enhancing robustness under high volatility, and achieving more balanced return–risk performance.

We propose **DBC-LSTM-ARIMA**—a *Dynamic Bi-directionally Coupled and Jointly-Optimized hybrid* for stock price prediction. The method upgrades the one-way residual-input architecture to a *two-way* coupling and aligns both components under a single window-level objective while retaining a light computational footprint. Our main contributions are:

- **How to turn a one-way ARIMA → LSTM pipeline into a bi-directionally coupled hybrid with feedback correction?** We keep the forward path that supplies ARIMA residuals to the LSTM, and add a feedback path that injects LSTM’s short-term prediction errors as exogenous corrections into ARIMA. This turns LSTM’s non-linear regime awareness into real-time adjustments of the linear component, reducing ARIMA’s systematic lag and bias during regime shifts.

- **How to coordinate the training objectives of ARIMA and LSTM within each walk-forward window?** Instead of training ARIMA and LSTM “each on its own,” we define a unified window-level loss that balances the two models’ errors and perform alternating updates—fix LSTM to refine ARIMA (including the feedback weight), then fix ARIMA to refine LSTM—until the joint objective stabilizes. This coordination mitigates parameter drift and reduces signal conflicts that often appear when components are tuned under different criteria.
- **How to adaptively fuse linear and non-linear forecasts under changing market regimes?** At the end of each window we combine the two point forecasts with a data-driven weight determined by recent error behavior (e.g., rolling error-ratio or a one-dimensional golden-section search over a convex weight). This lets the hybrid automatically tilt toward the more reliable component as the market rotates between “mostly linear” and “mostly non-linear” regimes, improving the return–volatility–drawdown profile. Across all steps we embed a non-anchored walk-forward optimization (training/validation/testing) and evaluate strategies under Long-Only and Long-Short rules with transaction costs, using standard portfolio diagnostics (annualized return and volatility, maximum drawdown, and modified information ratio).

The remainder of the paper is structured as follows. **Section II** surveys related work on ARIMA-only, LSTM-only, and ARIMA–LSTM hybrid strategies, positioning our study within this literature. **Section III** details the proposed DBC-LSTM-ARIMA methodology, including the bi-directional coupling, the window-level joint optimization, and the adaptive fusion mechanism under a walk-forward protocol. **Section IV** reports datasets, baselines, implementation details, and extensive experiments with ablations that isolate the benefits of feedback, joint training, and adaptive weighting, followed by portfolio back-tests and sensitivity analyses. **Section V** concludes with findings, limitations, and avenues for future research.

II. RELATED WORK

A. TIME SERIES MODELS

A considerable body of research has focused on forecasting stock prices using classical time series models. For instance, Karthik and Mohan [14] applied ARIMA to stock market data and demonstrated that the model is effective in capturing short-term linear patterns. Similarly, Pathak and Kapadia [15] employed ARIMA to analyze Indian stock indices and found that it could provide reliable forecasts when the data exhibits stable structures. Tang [16] explored the application of ARMA models for predicting closing prices of selected companies and confirmed their usefulness for short-horizon predictions with low error rates.

Beyond ARIMA, other econometric approaches have also been adopted. Recent studies show that Generalized Autoregressive Conditional Heteroskedasticity (GARCH) models remain standard tools for capturing volatility clustering in equity markets [17], [18]. Lin's line of work has been continued by more recent research: for example, Wang [19] uses GARCH-type models to describe and forecast the volatility of the Shanghai Composite Index and the Shenzhen Component Index, and discusses policy-relevant implications of the observed volatility dynamics. Contemporary empirical analyses also emphasize the role of Autoregressive Conditional Heteroskedasticity (ARCH) and GARCH processes in modeling equity-market volatility and assessing its impact on portfolio allocation and risk management. Furthermore, Teymurzade and Ślepaczuk [20] examines long-run U.S. stock-index data with ARIMA-based forecasting strategies, finding that although ARIMA models can perform relatively well during high-volatility episodes, their predictive performance tends to be weaker in calmer, more stable regimes.

To overcome the inherent limitations of single linear models, some scholars proposed combining statistical models with additional techniques. Jiang et al. [21] developed a composite approach integrating Principal Component Analysis (PCA) with Generalized Autoregressive Conditional Heteroskedasticity (GARCH) and deep networks, achieving improved predictive accuracy for stock prices. Pérez-Hernández et al. [22] and Michańkó et al. [23] enhanced GARCH-based volatility forecasts using neural-network architectures, showing that hybrid econometric-machine learning methods can better capture market dynamics. In addition, recent methodological studies and reviews [9], [24], [25] continue to emphasize that structural breaks and non-stationarity remain unresolved issues in financial time-series forecasting.

However, traditional time series models typically rely on strong assumptions such as linearity and stationarity, which are not always met in stock market data. Financial time series often display non-linear, non-stationary, and noisy behaviors, making it difficult for ARIMA or GARCH alone to deliver consistently accurate forecasts. As a result, while statistical models provide valuable insights into short-term dynamics and volatility patterns, their standalone predictive performance remains limited when applied to complex and rapidly changing markets.

B. MACHINE LEARNING MODELS

Over the past decade, machine learning has relaxed strict statistical assumptions and improved stock-index predictability by exploiting high-dimensional features such as technical indicators, volumes, and macro-signals. Classical learners—Support Vector Machines, Random Forests, Gradient Boosting, k-Nearest Neighbors, and Elastic-Net—have reported solid short-horizon gains when combined with careful feature engineering and cross-sectional enrichment [26], [27], [28], [29], [30]. Dimensionality reduction and representation

learning (PCA, auto-encoders) are often used to stabilize signals and suppress noise [31], [32], [33], [34]. Meta-learning and ensemble stacking further aggregate weak learners into more robust forecasters [35], [36].

Deep learning pushes this line by modeling long memory and complex temporal dependencies. Recurrent architectures (RNN, GRU and LSTM) improve sequence modeling for daily and intraday horizons [37], [38], [39], while 1-D CNNs extract local patterns from returns and order-flow features [40], [41]. Temporal Convolutional Networks and attention-based models (e.g., DA-RNN, Transformers) have shown competitive accuracy under regime variability by learning multi-scale dependencies and adaptive weights over time [42], [43], [44]. Hybrid representation strategies—such as combining CNN with LSTM for feature extraction and sequence prediction—also appear frequently in index-level studies [45], [46], [47].

Nevertheless, machine-learning forecasters face recurring pitfalls in financial time series: overfitting under volatility spikes, distribution shifts and concept drift, target leakage from improper cross-validation splits, and sensitivity to hyperparameters and transaction costs [48], [49], [50], [51]. Best practice therefore emphasizes walk-forward evaluation, regularization and early stopping, robust feature pipelines, and uncertainty-interpretability analyses (e.g., SHAP and LIME) to diagnose signal stability [52], [53], [54], [55]. These observations motivate methods that retain the interpretability and short-run structure of linear models while leveraging deep networks for non-linear residual structure—setting the stage for coupled or hybrid architectures considered next.

C. HYBRID MODELS

Hybrid methods integrate linear time-series structure with non-linear learners to exploit complementary inductive biases. A classic paradigm models the linear component with ARIMA and lets a non-linear learner fit the remainder, then combines both outputs; recent evidence with ARIMA–Artificial Neural Network (ARIMA–ANN) and related schemes reports consistent gains over single models on diverse financial and economic series [56], [57], [58], [59], [60]. Subsequent studies replace the non-linear block with Support Vector Machine (SVM), Random Forest (RF), Extreme Learning Machine (ELM) or gradient boosting and document improved short-horizon accuracy and robustness when noise filtering and walk-forward testing are used [61], [62], [63], [64].

A prominent line today couples ARIMA with LSTM. The most common pattern is residual modeling: ARIMA fits the short-run linear dynamics while an LSTM learns the non-linear remainder; forecasts are then summed [65], [66], [67], [68]. A closely related variant uses feature-level integration, where ARIMA residuals and market covariates are appended to the LSTM input and the LSTM produces the final signal—e.g., LSTM-ARIMA as a Hybrid Approach

in Algorithmic Investment Strategies, which evaluates the design under a non-anchored walk-forward protocol on major indices and reports risk-adjusted portfolio metrics [13]. Similar ARIMA-LSTM hybrids have been reported for equities, crypto, energy and traffic, generally outperforming either component alone under RMSE, MAE, or MAPE and simple trading backtests [69], [70].

A second stream precedes modeling with signal decomposition to mitigate non-stationarity, then learns each component with an appropriate learner before aggregating predictions. Typical decomposers include EMD/EEMD/CEEMDAN, wavelets and VMD [71], [72], [73], [74], [75]. Empirical studies such as EEMD-/CEEMDAN-LSTM, wavelet-ARIMA-LSTM and VMD-BiLSTM report noise reduction and improved short-term accuracy on stock-index series [76], [77], [78].

Despite solid progress, most ARIMA-LSTM hybrids share three structural limitations that motivate our work. First, information flow is typically one-directional (ARIMA to LSTM): ARIMA residuals are used as inputs to the LSTM, but there is no feedback from the deep model to refine ARIMA, as in Kashif and Ślepaczuk [13]. Second, the two parts are usually trained under separate objectives (e.g., AIC or MLE for ARIMA vs. MSE for LSTM) without a joint loss or coordinated updates within each walk-forward window, which risks objective mismatch and parameter drift [79], [80], [81]. Third, output fusion is often static (fixed or grid-searched weights) rather than regime-adaptive, leaving performance sensitive to market state changes [82], [83], [84]. These gaps motivate a hybrid that establishes bi-directional coupling, adopts a joint window-level objective with alternating updates, and deploys adaptive output fusion within a strict walk-forward framework—directions pursued in our DBC-LSTM-ARIMA.

D. SUMMARY

Taken together, prior research shows three complementary but imperfect streams. First, classical time-series models (e.g., ARIMA or GARCH) are effective at extracting short-run linear structure and volatility clustering, yet depend on stationarity and degrade around regime shifts. Second, machine-learning and deep-learning models (e.g., RF, SVM, GRU, LSTM, attention or Transformer variants) better capture non-linear and long-memory effects but are prone to overfitting, target leakage, and signal instability under volatility spikes. Third, hybrid designs—most notably ARIMA to LSTM pipelines that feed ARIMA residuals or features into an LSTM and evaluate by walk-forward testing—deliver practical gains but typically remain one-directional and separately trained, leaving room for bias accumulation and objective mismatch (see the representative LSTM-ARIMA study that appends ARIMA residuals to the LSTM input and uses the LSTM as the sole forecaster).

Guided by these observations, our methodology aims to retain the interpretability and linear strengths of ARIMA

while leveraging LSTM's non-linear capacity, and to overcome the limitations of one-way, independently trained hybrids. In the next section we introduce a dynamic, bi-directionally coupled and jointly optimized ARIMA-LSTM framework with adaptive output fusion, embedded in a rigorous walk-forward process. This design targets improvements in out-of-sample accuracy and risk-adjusted performance while keeping computational costs acceptable for algorithmic investment use.

III. AN INNOVATIVE APPROACH: DBC-LSTM-ARIMA

DBC-LSTM-ARIMA is a dynamic, bi-directionally coupled and jointly-optimized hybrid designed for stock price prediction under a strict walk-forward protocol. The model structure is shown in Fig. 1. In contrast to one-way hybrids—where ARIMA residuals are appended to an LSTM and the final signal is produced solely by the LSTM—our design establishes two information paths and a single window-level objective. (A representative one-way formulation, LSTM-ARIMA as a Hybrid Approach in Algorithmic Investment Strategies, feeds ARIMA residuals and covariates to the LSTM and uses the LSTM output for trading; we adopt the same practical walk-forward setup as a strong baseline. Concretely, within each non-anchored walk-forward window (training/validation/testing), we first fit an ARIMA to the differenced series, choose orders by an information criterion, and compute one-step forecasts and residuals. These residuals, together with a history window of prices and optional market covariates (e.g., volatility and volume), constitute the forward coupling that enters the LSTM. After training the LSTM on the same window, we form the reverse coupling: the LSTM's one-step prediction errors over the window are injected as an exogenous correction into the ARIMA (ARIMAX-style), with a learnable feedback weight. Both model families are then brought under a joint window-level loss that balances their errors. We optimize this objective by alternating updates: fix LSTM parameters and refine ARIMA (including the feedback weight) by gradient-based steps around the likelihood solution; then fix ARIMA and update LSTM with standard back-propagation; repeat until the joint loss stalls by a tolerance or early-stopping rule. This loop ties the linear and non-linear components to the same target, mitigating objective mismatch and curbing parameter drift across windows. At inference time, we produce two point forecasts—refined ARIMA and LSTM—and adaptively fuse them with a convex weight determined from recent window errors (e.g., a rolling error-ratio rule or a 1-D golden-section search over the weight). The resulting hybrid forecast directly drives Long-Only or Long-Short trading with explicit transaction costs, while standard portfolio diagnostics (annualized return or volatility, drawdown, modified information ratio) are tracked window by window. The architecture keeps computation modest—orders for ARIMA remain small, the LSTM uses one to two hidden layers with dropout, and all optimization is confined to the current window—yet it captures regime shifts through feedback reconstruction, aligns

both components through a unified objective, and adapts forecast blending to prevailing market conditions, thereby targeting higher out-of-sample accuracy and more stable risk-adjusted performance than one-directional or separately trained hybrids.

A. THE SLIDING-WINDOW PREPROCESSING MODULE

Let the univariate price (or log-price) series be $\{y_t\}_{t=1}^T$. Fix a window length W for walk-forward optimization, and let n denote the inner iteration index used later by the coupling loop (initialized as $n = 0$).

1) WINDOW CONSTRUCTION

The series is segmented into overlapping windows of length W . Denote by t_r the starting time of the r -th window, with $t_0 = 1$ and $t_{r+1} = t_r + 1$. The number of windows is

$$R = T - W + 1. \quad (1)$$

For each window we extract the raw segment.

$$Y_{t_r} = [y_{t_r}, y_{t_r+1}, \dots, y_{t_r+W-1}], r = 0, 1, \dots, R. \quad (2)$$

This **non-anchored sliding scheme**, which moves the window forward one step at a time without tying it to any fixed anchor point, guarantees that every observation is used in exactly one position per window and that the training, validation, and testing splits (defined later) respect temporal order.

2) STATIONARIZATION BY DIFFERENCING AND ADF TESTING

Financial prices are typically non-stationary. Before fitting the linear component we difference each window and test for stationarity. Let Δ denote the backward difference operator, $\Delta y_t = y_t - y_{t-1}$. Its d -fold application produces $\Delta^d y_t = (1 - L)^d y_t$, where L is the lag operator. For window r , we choose the smallest integer $d \geq 0$ such that the Augmented Dickey-Fuller (ADF) test rejects the unit-root hypothesis on the training fold of Y_{t_r} at a preset level (e.g., 1% or 5%). The resulting stationary window series is

$$Z_{t_r} = \Delta^d Y_{t_r}, r = 0, 1, \dots, R. \quad (3)$$

We store the adopted for each window so that subsequent ARIMA forecasts can be **inverse-differenced** back to the original scale in the prediction stage.

B. THE INITIALIZING ARIMA ERROR MODULE

This module models the linear component in each sliding window and initializes the ARIMA prediction error, which will be used in subsequent coupling and joint optimization.

For the r -th window, we denote the stationary series as $Z_{t_r} = \Delta^d Y_{t_r}$ (see Section III-A). Within the window, t indexes time, and the window index is $r = 0, 1, \dots, R$.

1) ORDER SELECTION: AIC CRITERION

Given the differencing order d , we select the optimal combination (p, q) via grid search. For each candidate (p, q) ,

model parameters are estimated (see below), and the Akaike Information Criterion (AIC) is computed:

$$\text{AIC} = 2k - 2l, \quad (4)$$

where k is the number of parameters and l is the log-likelihood value. The (p, d, q) with the smallest AIC is chosen as the ARIMA order for the current window.

2) PARAMETER ESTIMATION: MAXIMUM LIKELIHOOD (MLE)

Within window r , the ARIMA (specifically, ARMA(p, q) on Z_{t_r}) is written as

$$z_{t_r} = \sum_{i=1}^p \phi_i z_{t_r-i} + \sum_{j=1}^q \theta_j \varepsilon_{t_r-j} + \mu + \varepsilon_{t_r}, r = 0, 1, \dots, R \quad (5)$$

where $\{\varepsilon_{t_r-j}\}, j = 1, 2, \dots, q$ and ε_{t_r} are the innovation term with zero mean and constant variance. Using exact likelihood or innovation algorithms (e.g., Kalman filtering), we maximize the log-likelihood to obtain estimates $\{\hat{\phi}_i\}, \{\hat{\theta}_j\}, \hat{\mu}$, from which one-step forecasts $\hat{z}_{t_r}$ and innovations $\hat{\varepsilon}_{t_r}$ are derived.

3) INVERSE DIFFERENCING: RESTORING FORECASTS TO THE ORIGINAL SCALE

The one-step forecasts in the differenced scale are mapped back to the price scale through inverse differencing, yielding ARIMA predictions:

$$\hat{y}_{t_r}^{\text{ARIMA}} = \text{InverseDiff}(z_{t_r}), r = 0, 1, \dots, R. \quad (6)$$

Here $\text{InverseDiff}(\cdot)$ uses the d anchor points from the beginning of Y_{t_r} to restore predictions consistently to the original scale.

4) INITIALIZATION OF ARIMA PREDICTION ERROR

The initial error (iteration counter $n = 0$) for window r is defined as the difference between the observed price and ARIMA prediction:

$$e_{t_r}^{(\text{ARIMA}, 0)} = y_{t_r} - \hat{y}_{t_r}^{\text{ARIMA}}, r = 0, 1, \dots, R. \quad (7)$$

Through these steps, each sliding window obtains ARIMA orders consistent with AIC, MLE-estimated parameters, and original-scale forecasts with initialized errors, thus providing a rigorous and reproducible linear foundation for the bi-directional coupling and alternating optimization to follow.

C. THE FORWARD COUPLING MODULES

This module trains the LSTM inside each sliding window by feeding ARIMA information forward. It converts the linear model's unexplained part into features for the non-linear learner and produces the LSTM's window-level prediction and error for later feedback.

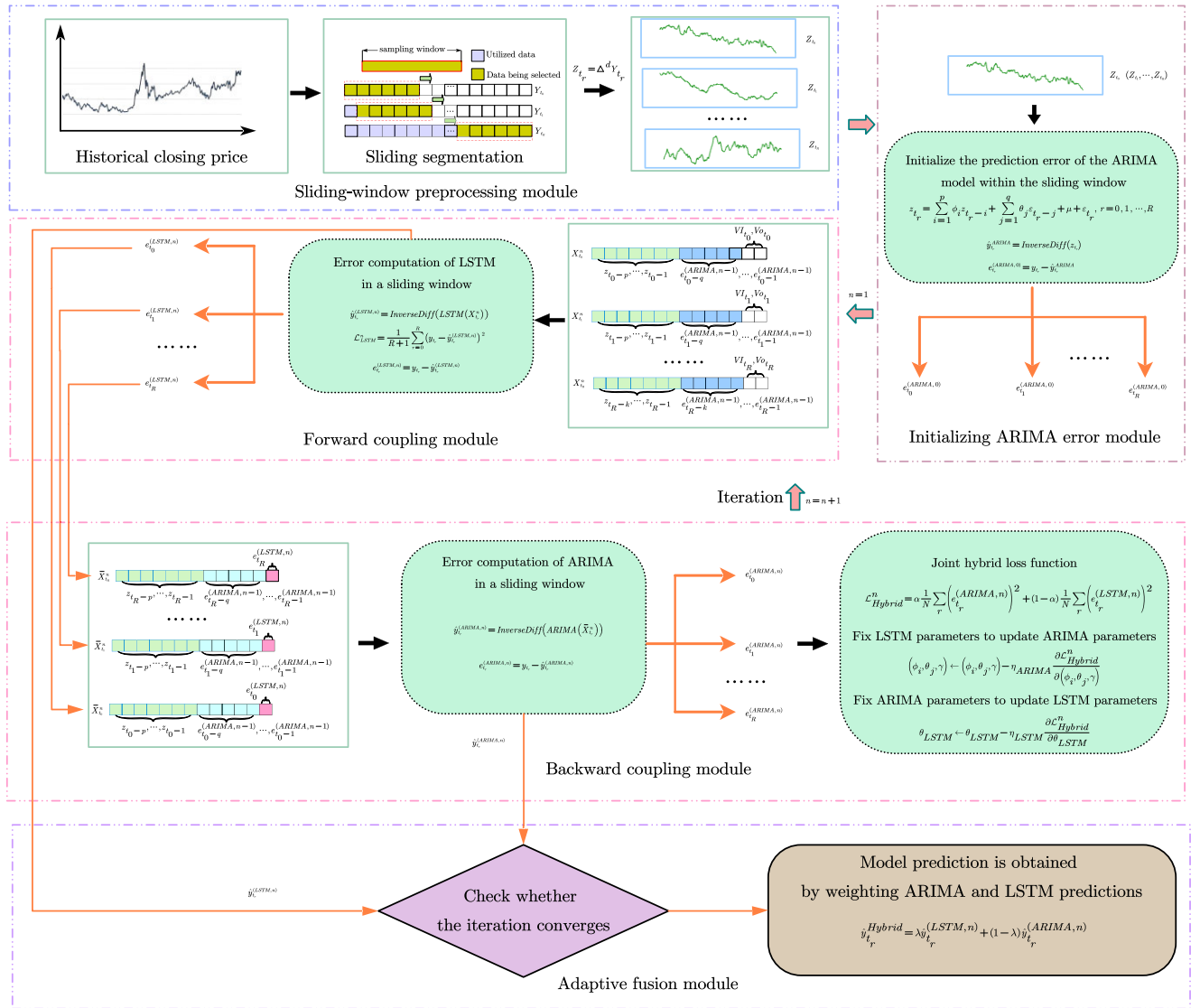


FIGURE 1. The structure of method DBC-LSTM-ARIMA, including the sliding-window preprocessing module, initializing ARIMA error module, forward and backward coupling modules, and adaptive fusion.

1) ITERATION UPDATE

At the beginning of the window-internal loop, increase the iteration counter:

$$n = n + 1. \quad (8)$$

2) FEATURE CONSTRUCTION FOR FORWARD COUPLING

For the current window r we build a feature vector that concatenates (i) a history of the stationary price differences, (ii) a history of ARIMA residuals from the previous iteration $n-1$, and (iii) auxiliary market descriptors:

$$X_r^n = [z_{t_r-p}, \dots, z_{t_r-1}, e_{t_r-q}^{(ARIMA,n-1)}, \dots, e_{t_r-1}^{(ARIMA,n-1)}, V I_r, V O_r], \quad (9)$$

where $r = 0, 1, \dots, R$. Here p is the length of the closing-price history window $[z_{t_r-p}, \dots, z_{t_r-1}]$; q is the

number of ARIMA-residual lags used from the previous iteration $[e_{t_r-k}^{(ARIMA,n-1)}, \dots, e_{t_r-1}^{(ARIMA,n-1)}]$; $V I_r$ and $V O_r$ denote optional auxiliary signals such as volatility and volume aligned at time t_r .

3) LSTM MAPPING

Given X_r^n , the LSTM produces the one-step prediction of the raw price at the same alignment index:

$$\hat{y}_r^{(LSTM,n)} = \text{InverseDiff}(LSTM(X_r^n)), \quad r = 0, 1, \dots, R. \quad (10)$$

$\text{InverseDiff}(\cdot)$ uses the d anchor points from the beginning of $LSTM(X_r^n)$ to restore predictions consistently to the original scale. The network can be a single- or double-layer LSTM with dropout; weights are confined to the current window to preserve the walk-forward protocol.

4) WINDOW-LEVEL LSTM LOSS (MSE)

For iteration n , the LSTM is trained by minimizing the mean-squared error over all windows:

$$\mathcal{L}_{LSTM}^n = \frac{1}{R+1} \sum_{r=0}^R \left(y_{t_r} - \hat{y}_{t_r}^{(LSTM,n)} \right)^2. \quad (11)$$

This definition aligns the non-linear component with the same prediction target used by the linear component, preparing both for the joint objective in the backward-coupling stage.

5) LSTM PREDICTION ERROR

We record the LSTM's one-step error for each window:

$$e_{t_r}^{(LSTM,n)} = y_{t_r} - \hat{y}_{t_r}^{(LSTM,n)}, r = 0, 1, \dots, R. \quad (12)$$

The sequence $e_{t_r}^{(LSTM,n)}$ is the backward-coupling signal used in the next module to adjust the ARIMA part (as an exogenous correction), closing the bi-directional loop.

D. THE BACKWARD COUPLING MODULES

This module implements the feedback reconstruction of the linear component inside each sliding window by injecting the LSTM's one-step error as an exogenous correction. It closes the bi-directional loop and produces the updated ARIMA prediction and error for iteration.

From Section III-C we obtain, for every window r , the LSTM one-step error is shown as the formula (12). We also keep the ARIMA residual history from the previous iteration, $[e_{t_r-q}^{(ARIMA,n-1)}, \dots, e_{t_r-1}^{(ARIMA,n-1)}]$, and the differenced price history $[z_{t_r-p}, \dots, z_{t_r-1}]$.

1) FEATURE CONSTRUCTION FOR BACKWARD COUPLING

We stack the above into the backward-coupling input vector:

$$\bar{X}_{t_r}^n = \left[z_{t_r-p}, \dots, z_{t_r-1}, e_{t_r-q}^{(ARIMA,n-1)}, \dots, e_{t_r-1}^{(ARIMA,n-1)}, e_{t_r}^{(LSTM,n)} \right], \quad (13)$$

where $r = 0, 1, \dots, R$. The first two blocks mirror the standard AR and MA lags used by the ARIMA model within window r ; the last entry $e_{t_r}^{(LSTM,n)}$ is the feedback signal that conveys the portion of the series not explained by the linear component at iteration $n-1$.

2) FEEDBACK-RECONSTRUCTED ARIMA MODEL

On the stationary scale of the window we reconstruct ARIMA by adding the LSTM error as an exogenous term:

$$\hat{y}_{t_r}^{(ARIMA,n)} = \text{InverseDiff} \left(\text{ARIMA} \left(\bar{X}_{t_r}^n \right) \right) \quad (14)$$

$$\text{ARIMA} \left(\bar{X}_{t_r}^n \right) = \sum_{i=1}^p \phi_i z_{t-i} + \sum_{j=1}^q \theta_j e_{t-j}^{(ARIMA,n-1)} + \mu + \gamma e_{t_r}^{(LSTM,n)} \quad (15)$$

where $r = 0, 1, \dots, R$ and parameters are estimated by maximum likelihood. Here $\{\phi_i\}_{i=1}^p$ and $\{\theta_j\}_{j=1}^q$ denote the AR and

MA coefficients of the linear part on z_{t-i} and $e_{t-j}^{(ARIMA,n-1)}$; μ is the intercept; and γ weights the feedback term $e_{t_r}^{(LSTM,n)}$. The likelihood is computed on the training fold of window γ using the likelihood (e.g., Kalman filter), with standard stability constraints on ϕ_i and θ_j . The mapping ARIMA $(\cdot)$ includes inverse differencing with the window's stored d so that $\hat{y}_{t_r}^{(ARIMA,n)}$ is expressed on the original price scale, consistent with earlier steps.

3) FEEDBACK ARIMA ERROR

Using the observed price y_{t_r} , we compute the updated ARIMA model error at iteration n :

$$e_{t_r}^{(ARIMA,n)} = y_{t_r} - \hat{y}_{t_r}^{(ARIMA,n)}, r = 0, 1, \dots, R \quad (16)$$

4) JOINT HYBRID LOSS OPTIMIZATION AND PARAMETER UPDATES

Having obtained the feedback ARIMA prediction and error, as well as the LSTM prediction and error from Section 3.3, we optimize both parts within the same window by a joint objective and alternating updates.

① Hybrid loss

Define the mixed objective that aggregates errors from both components:

$$\mathcal{L}_{Hybrid}^n = \alpha \frac{1}{N} \sum_{r=0}^R \left(e_{t_r}^{(ARIMA,n)} \right)^2 + (1 - \alpha) \frac{1}{N} \sum_{r=0}^R \left(e_{t_r}^{(LSTM,n)} \right)^2 \quad (17)$$

with equal weighting $\alpha = 0.5$ by default. Here N is the number of samples aggregated at the window level (typically $N = R+1$), and the sums run over $r = 0, 1, \dots, R$. This loss ties the linear and non-linear branches to a single optimization target.

② Update ARIMA and feedback parameters

Fix the LSTM parameters and refine the linear branch-AR and MA coefficients and the feedback weight-by descending the gradient of the hybrid loss:

$$(\phi_i, \theta_j, \gamma) \leftarrow (\phi_i, \theta_j, \gamma) - \eta_{ARIMA} \frac{\partial \mathcal{L}_{Hybrid}^n}{\partial (\phi_i, \theta_j, \gamma)}. \quad (18)$$

Practically, we initialize ϕ_i, θ_j, μ at the MLE solution from Section III-D(2), keep the stability constraints, and treat γ as a learnable scalar that scales the exogenous term $e_{t_r}^{(LSTM,n)}$. One or several small gradient steps are taken on the training fold of the window.

③ Update LSTM parameters

In this step we fix the ARIMA-side parameters $(\phi_i, \theta_j, \mu, \gamma)$ and update the LSTM under the same joint objective used for the linear branch. The parameter update follows a single gradient step with respect to the hybrid loss:

$$\theta_{LSTM} \leftarrow \theta_{LSTM} - \eta_{LSTM} \frac{\partial \mathcal{L}_{Hybrid}^n}{\partial \theta_{LSTM}} \quad (19)$$

Here η_{LSTM} denotes the learning rate. This update enforces coordinated learning by driving the LSTM to minimize the joint window-level objective, after which we return to the ARIMA update in (4.2) and continue alternating until convergence.

④ Feedback iteration

Collect the ARIMA errors of iteration n across all windows and form the stack

$$e^{(ARIMA,n)} = \left[e_{t_0}^{(ARIMA,n)}, e_{t_1}^{(ARIMA,n)}, \dots, e_{t_R}^{(ARIMA,n)} \right]^\top. \quad (20)$$

These errors become the **residual history** used by the next forward-coupling step. For each window, the feature vector for iteration is updated by replacing the ARIMA-residual lags with the new ones:

$$X_{t_r}^{n+1} = \left[z_{t_r-p}, \dots, z_{t_r-1}, e_{t_r-q}^{(ARIMA,n)}, \dots, e_{t_r-1}^{(ARIMA,n)}, V I_{t_r}, V O_{t_r} \right]. \quad (21)$$

Finally, advance the iteration counter and continue the loop:

$$n \leftarrow n + 1.$$

If a convergence rule is employed—e.g., the change in the hybrid loss falls below a tolerance—terminate; otherwise return to Section III-C with the updated residual stack.

E. THE ADAPTIVE FUSION MODULE

After the bidirectional coupling and the joint optimization in the current sliding window have converged (Sections III-C-III-D), we obtain two one-step forecasts at the same alignment index t_r : the LSTM output $\hat{y}_{t_r}^{(LSTM,n)}$ and the feedback-reconstructed ARIMA output $\hat{y}_{t_r}^{(ARIMA,n)}$. The final forecast is the convex fusion of these two paths:

$$\hat{y}_{t_r}^{\text{Hybrid}} = \lambda \hat{y}_{t_r}^{(LSTM,n)} + (1 - \lambda) \hat{y}_{t_r}^{(ARIMA,n)}, r = 0, 1, \dots, R. \quad (22)$$

Here $\lambda \in [0, 1]$ is the fusion weight (the weight of the LSTM path, while $1 - \lambda$ is the weight of the ARIMA path). We estimate λ within the same window using only the validation fold to avoid look-ahead.

The golden-section search for λ is described as follows.

For the r -th sliding window, $Y_{t_r} = [y_{t_r}, y_{t_r+1}, \dots, y_{t_r+W-1}]$. Let $\mathcal{V}_r = \{t_r, \dots, t_r + W - 1\}$. For any index $t \in \mathcal{V}_r$, y_t denotes the observed price at time t ; $\hat{y}_{t_r}^{(LSTM,n)}$ and $\hat{y}_{t_r}^{(ARIMA,n)}$ are the one-step forecasts produced in Sections III-C and III-D at the same time t .

We choose $\lambda \in [0, 1]$ to minimize the validation MSE of the fused forecast:

$$f_r(\lambda) = \frac{1}{N_{\mathcal{V}_r}} \sum_{t \in \mathcal{V}_r} \left(y_t - \lambda \hat{y}_{t_r}^{(LSTM,n)} - (1 - \lambda) \hat{y}_{t_r}^{(ARIMA,n)} \right)^2. \quad (23)$$

Let $\varphi = (\sqrt{5} - 1)/2 \approx 0.618$. Initialize

$$[a, b] = [0, 1], c = b - \varphi(b - a), d = a + \varphi(b - a) \quad (24)$$

and evaluate $f_r(c), f_r(d)$.

Iterate until the interval length is below a tolerance ε (e.g., 10^{-3}):

If $f_r(c) \leq f_r(d)$: set $b \leftarrow d, d \leftarrow c, c \leftarrow b - \varphi(b - a)$; compute $f_r(c)$.

Else: set $a \leftarrow c, c \leftarrow d, d \leftarrow a + \varphi(b - a)$; compute $f_r(d)$.

When $b - a < \varepsilon$, take $\lambda_r^* = (a + b)/2$ (or equivalently choose the better of the last c and d). The selected λ_r^* (validation-only) is then used in the fusion

$$\hat{y}_{t_r}^{\text{Hybrid}} = \lambda_r^* \hat{y}_{t_r}^{(LSTM,n)} + (1 - \lambda_r^*) \hat{y}_{t_r}^{(ARIMA,n)}, r = 0, 1, \dots, R, \quad (25)$$

for the test step (and subsequently for trading) of the same window. Because $f_r(\lambda)$ is a convex quadratic in λ with fixed forecasts, the golden-section search converges rapidly and is numerically stable.

IV. EXPERIMENT

A. DATASET

In this study, we employ three representative A-share stocks—BYD (002594.SZ), China Unicom (600050.SS), and Ping An Bank (000001.SZ)—as experimental datasets to evaluate the predictive efficacy of the proposed DBC-LSTM-ARIMA model. The pertinent data is presented in Table 1. The sample spans September 1, 2011 to September 1, 2025 at a daily frequency. For each stock we retrieve the standard price-volume fields, namely daily opening price (OPEN), high price (HIGH), low price (LOW), forward-adjusted closing price (CLOSE), and trading volume (VOLUME). Forward adjustment ensures continuity around corporate actions (e.g., splits, dividends) while leaving intraday range variables (OPEN, HIGH and LOW) and VOLUME unchanged. The datasets were obtained through the Tushare financial data interface and aligned to trading days on the Shanghai and Shenzhen exchanges.

Fig. 2. illustrates the rolling evaluation protocol adopted in this study. Each row corresponds to one walk (top to bottom: Walk 1 $\rightarrow$ Walk 8). Within every walk, the training segment (green, 1000 trading days), validation segment (yellow, 250 trading days), and out-of-sample test segment (red, 250 trading days) are contiguous; the step size equals the test length (250 trading days), so the window slides forward by one test block each time. The dark-red bars at the bottom depict the trajectory of full WFO windows across the sample, clarifying the coverage and non-overlap of OOS periods. This non-anchored scheme (1000/250/250, step=250) is used consistently for all assets and all experiments in the paper.

The plots in **Fig. 3.** report daily CLOSE series aligned on a common trading calendar: BYD (002594.SZ), China Unicom (600050.SS), and Ping An Bank (000001.SZ). BYD exhibits a prolonged upward drift with pronounced volatility

TABLE 1. Statistics of the three datasets.

Stocks	Stock code	Stock sector	Industry
BYD	002594.SZ	Shenzhen A-share	Manufacturing
China Unicom	600050.SS	Shanghai A-share	Communication
Ping An Bank	000001.SZ	Shenzhen A-share	Finance

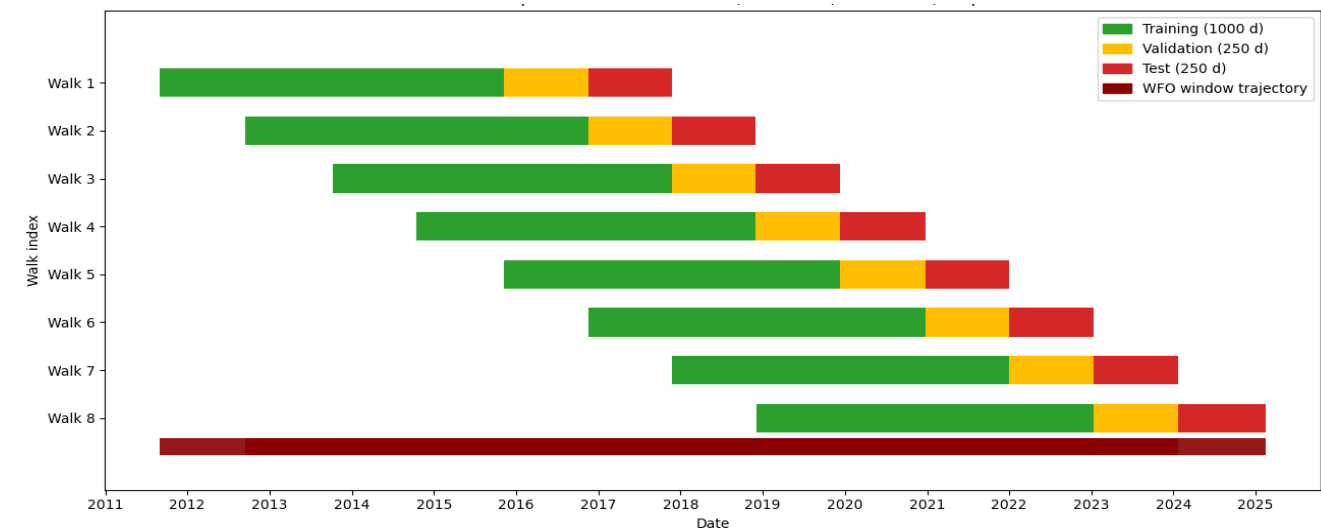


FIGURE 2. Walk-forward optimization (non-anchored) with ~5-years In-Sample (IS) and 1-year Out-of-Sample (OOS). **Note:** Green bars denote the training segment, yellow the validation segment, and red the out-of-sample test segment;dark-red bars trace the WFO window trajectory across the full sample. The experiment uses daily A-share data for BYD(002594.SZ), China Unicom(600050.SS), and Ping An Bank(000001.SZ) from 2011-09-01 to 2025-09-01. The training window is 1000 trading days, and the validation and testing windows are 250 trading days each; the step size equals 250 trading days.

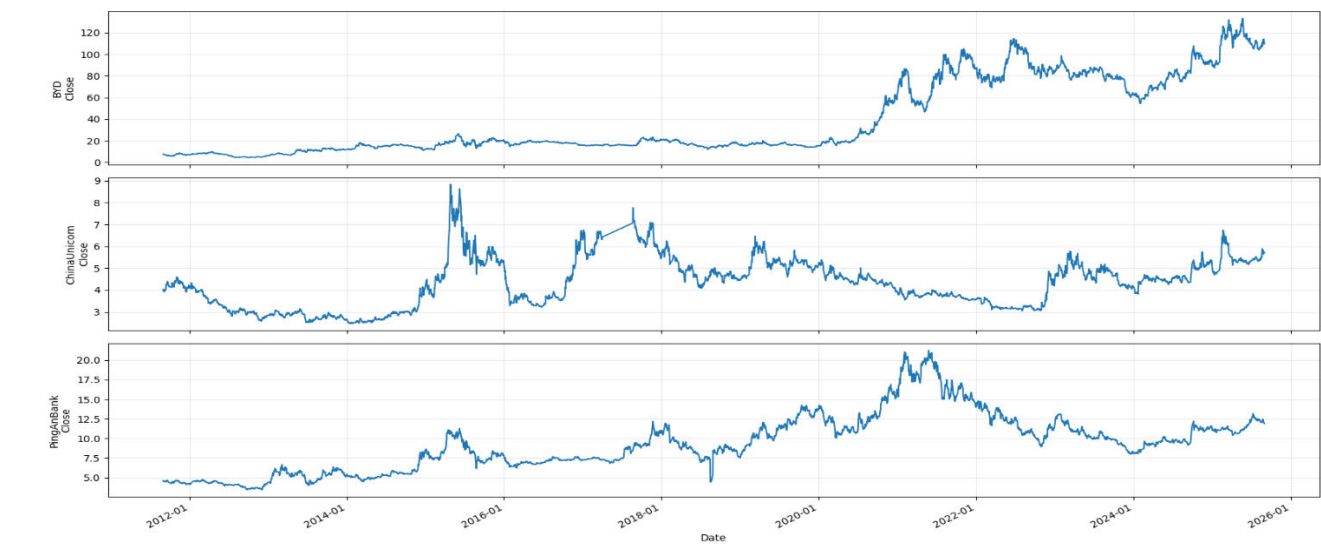


FIGURE 3. BYD, China Unicom and Ping an back closing price trend chart.

clusters—especially the strong rally from 2020 onward—typical of growth-oriented manufacturing. China Unicom shows episodic spikes (e.g., 2015–2017) followed by mean-reverting phases, reflecting the telecommunication sector’s

policy- and event-driven dynamics. Ping An Bank displays cyclical behavior with sizeable drawdowns (e.g., 2018, pandemic shock) and subsequent recoveries, representative of financials. The three series jointly contain non-smoothness,

non-stationarity, and regime shifts, motivating the use of a hybrid, bi-directionally coupled model that leverages linear structure while adapting to non-linear variation.

B. EVALUATION METRICS FOR PRICE PREDICTION

For the r -th walk, let $\mathcal{T}_r \subset \{t_r, \dots, t_r + W - 1\}$ denote the OOS test index set, and let $m_r = |\mathcal{T}_r|$. For each $t \in \mathcal{T}_r$, y_t is the true closing price and $\hat{y}_t^{Hybrid}$ is the corresponding fused prediction (Section III-E). Define the test-set mean $\bar{y}_r = \frac{1}{m_r} \sum_{t \in \mathcal{T}_r} y_t$. We report the following metrics per walk:

$$MAE_r = \frac{1}{m_r} \sum_{t \in \mathcal{T}_r} |y_t - \hat{y}_t^{Hybrid}| \quad (26)$$

$$RMSE_r = \sqrt{\frac{1}{m_r} \sum_{t \in \mathcal{T}_r} (y_t - \hat{y}_t^{Hybrid})^2} \quad (27)$$

$$MAPE_r = \frac{1}{m_r} \sum_{t \in \mathcal{T}_r} \left| \frac{y_t - \hat{y}_t^{Hybrid}}{y_t} \right| \quad (28)$$

$$R_r^2 = 1 - \frac{\sum_{t \in \mathcal{T}_r} (y_t - \hat{y}_t^{Hybrid})^2}{\sum_{t \in \mathcal{T}_r} (y_t - \bar{y}_r)^2} \quad (29)$$

All metrics are computed only on the OOS test segment $\mathcal{T}_r$ of each nonanchored WFO walk (Section IV-A). We report both per-walk results and aggregate statistics. Lower MAE_r / $RMSE_r$ / $MAPE_r$, together with higher R_r^2 indicate better predictive performance.

C. EXPERIMENT SETTINGS

We run DBC-LSTM-ARIMA under a non-anchored walk-forward optimization (WFO) protocol. Experiment settings are shown in Table 2. Each walk uses 1000/250/250 trading days for the training/validation/test segments, respectively, and the window is shifted forward by 250 trading days, yielding eight walks with non-overlapping OOS test blocks (total window length $W = 1500$ trading days). In the sliding-window preprocessing module, the differencing order $d \in \{0, 1\}$ for each walk is selected on the training split by the Augmented Dickey-Fuller (ADF) test at the 5% level; if ADF indicates stationarity we set $d = 0$, otherwise $d = 1$ (in our sample this rule yields $d = 1$). In the initializing ARIMA error module, ARIMA orders (p, q) are chosen on the training split by a grid search over $p, q \in \{0, 1, 2, 3, 4, 5\}$, selecting the pair that minimizes AIC (ties resolved by smaller q , then smaller p); the parameters (ϕ_i, θ_j, μ) are then estimated on the training split by Gaussian maximum likelihood using an exact innovations/state-space likelihood with an L-BFGS optimizer (maximum 500 iterations, tolerance 10^{-6}).

For the forward-coupling module, the nonlinear branch is a single-layer LSTM with 64 hidden units and tanh activation followed by a linear head, with dropout 0.1. It is trained per walk with mean-squared-error loss using Adam (learning rate 1×10^{-3}), batch size 64, at most 20 epochs, and early stopping (patience 5, minimum learning rate 1×10^{-5}); the final LSTM for each walk is selected by the lowest validation

MSE (random seed 2025). In the backward-coupling module, the AR and MA orders (p, q) are fixed to those selected in the initializing ARIMA error module. Starting from the MLE solution for (ϕ_i, θ_j, μ) , we initialize the feedback weight $\gamma = 0$ and update $(\phi_i, \theta_j, \mu, \gamma)$ by minimizing the hybrid loss with learning rate $\eta_{ARIMA} = 1 \times 10^{-3}$, performing one gradient (or quasi-Newton) step on the training split per alternation cycle. The hybrid loss is the window-averaged mixture of ARIMA and LSTM squared errors with weight 0.5 (robustness checked for weights in $[0.3, 0.7]$). We alternate forward $\leftrightarrow$ backward updates for at most 10 cycles, or until the relative decrease in the hybrid loss falls below 10^{-4} for three consecutive cycles.

Finally, in the adaptive-fusion module we determine the fusion weight $\lambda \in [0, 1]$ inside each walk on the validation split via golden-section search (objective: validation MSE; tolerance 10^{-3}), and keep the selected λ fixed for the corresponding OOS test segment.

D. ANALYSIS OF EXPERIMENTAL RESULTS

To assess predictive performance, we train DBC-LSTM-ARIMA separately on the three A-share stocks (BYD, China Unicom, Ping An Bank) under the non-anchored WFO scheme described above. After finishing the window-internal alternating optimization for each walk, the model is evaluated only on the out-of-sample test segments. Table 3 summarizes the test-set ranges for the three assets across walks. The learning dynamics of our model—i.e., the evolution of the training/validation loss within each window—are visualized in Fig. 4. The out-of-sample prediction results on the three datasets are illustrated in Figs. 5–7 (corresponding to the respective stocks), where we plot the predicted close price against the realized price. These figures provide an intuitive view of the fit quality before we proceed to quantitative evaluation with the price-level and strategy-level metrics.

Notably, from Fig. 5 we observe that DBC-LSTM-ARIMA performs particularly well on the Ping An Bank dataset. As a representative financial stock, Ping An Bank's closing price exhibits a comparatively stable trend with moderate volatility, which reduces the burden on the non-linear learner and allows the linear ARIMA part to capture short-run structure effectively. The bi-directional coupling then makes only small, targeted corrections, yielding tight tracking of the test series and limited error bursts at regime edges. The strong out-of-sample fit on Ping An Bank indicates that the proposed hybrid enjoys high robustness and accuracy when confronted with equities whose price dynamics are trend-dominated and less erratic, corroborating the intended advantage of combining a linear backbone with lightweight non-linear feedback and adaptive fusion.

Moreover, from Fig. 6 we can see that DBC-LSTM-ARIMA also performs well on the BYD dataset. BYD is a representative manufacturing stock whose forward-adjusted closing price exhibits large fluctuations and intermittent spikes, posing a greater challenge to forecasting than the relatively stable financial series. Even under this high-volatility,

TABLE 2. Experiment settings.

Module	Item	Value or range	Setting
Sliding-window preprocessing module	Window length W	Train 1000 d + Val 250 d + Test 250 d (total $W=1500$ trading days)	Non-anchored WFO
	Step size	250 trading days	Each walk shifts forward by one test block.
	Total #windows R	8	Computed on the common trading calendar of the three assets using $R = \lfloor (T - W) / \text{Step} \rfloor + 1$.
	Differencing order d	$d \in \{0, 1\}$	Selected on the training split of each walk by ADF (5% level). If ADF indicates stationarity, set $d = 0$; in our sample the rule yields $d = 1$.
Initializing ARIMA error module	ARIMA orders (p, q)	$p, q \in \{0, 1, 2, 3, 4, 5\}$	Grid on the training split; choose the pair that minimizes AIC (ties resolved by smaller q , then smaller p).
	Parameter estimation (ϕ_i, θ_j, μ)	Estimated using	Gaussian MLE on the training split (exact innovations/state-space likelihood); optimizer L-BFGS, max 500 iterations, tolerance $1e-6$.
Forward coupling module	LSTM backbone	$1 \times \text{LSTM}$ (64 hidden units, tanh) $\rightarrow$ linear head; Dropout 0.1	Fixed architecture across assets and walks.
	Training setup	Loss: MSE; Optimizer: Adam (lr 1×10^{-3}); batch 64; max 20 epochs; early stop (patience 5, min-lr 1×10^{-5})	Model selection by validation MSE; seed 2025 for reproducibility.
Backward coupling module	AR/MA orders (p, q)	inherited per window	Fixed to the (p, q) selected in the initializing ARIMA error module (AIC on the training split).
	Parameters updated $(\phi_i, \theta_j, \mu, \gamma)$	Estimated using	Start from the MLE solution of Section 3.2 for ϕ_i, θ_j, μ ; initialize feedback weight $\gamma = 0$. All are learned in the backward step.
	Optimizer / step size for ARIMA side	learning rate $\eta_{\text{ARIMA}} = 1 \times 10^{-3}$	One gradient (or quasi-Newton) update on the training split per alternation cycle.
	Hybrid loss weight	0.5	Fixed (robustness checked in $[0.3, 0.7]$); loss is the window-averaged mixture of ARIMA and LSTM squared errors.
	Alternation stopping rule max 10 cycles or relative loss change $< 10^{-4}$ (per window)	for 3 consecutive cycles.	Applied to the joint hybrid loss to terminate the forward $\leftrightarrow$ backward loop.
Adaptive fusion module	Fusion weight λ	$[0, 1]$	Estimated inside the window on the validation split by golden-section search (objective: validation MSE; tolerance 10^{-3}).

TABLE 3. Test-set ranges of the three datasets.

Stocks	Data time range	Training	Verification	Test	Ratio
BYD	September 1, 2011 – September 1, 2025	1000	250	250	4:1:1
China Unicom	September 1, 2011 – September 1, 2025	1000	250	250	4:1:1
Ping An Bank	September 1, 2011 – September 1, 2025	1000	250	250	4:1:1

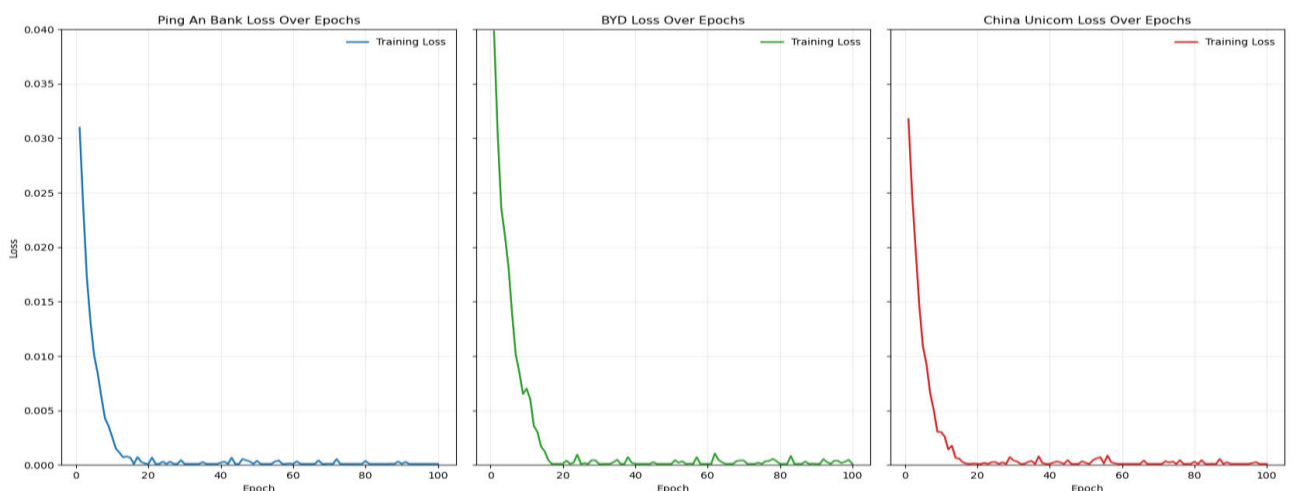


FIGURE 4. Change chart of training loss of three data sets.

non-stationary regime, the hybrid model tracks the test curve closely, with only small deviations around the sharpest jumps.

This behavior is consistent with the design of our approach: the linear ARIMA backbone captures short-run structure

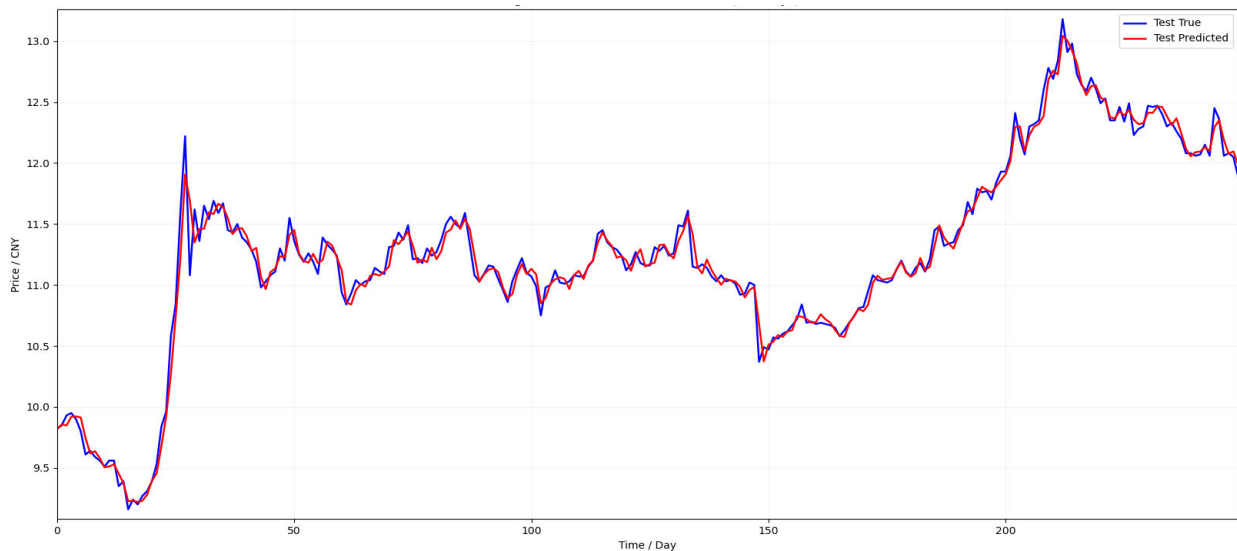


FIGURE 5. Comparison between the predicted prices and the truth prices of stock Ping An Bank.

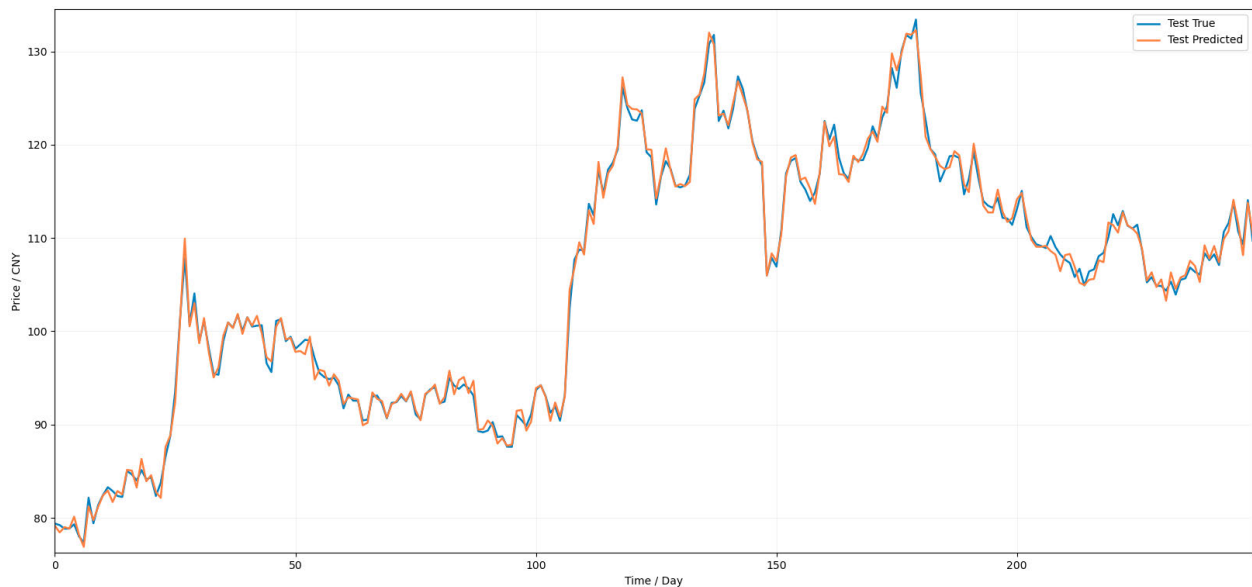


FIGURE 6. Comparison between the predicted prices and the truth prices of stock BYD.

while the backward feedback and adaptive fusion provide timely non-linear corrections. Taken together, the results on BYD indicate that the proposed method maintains robust prediction accuracy even when price dynamics are highly variable.

From Fig. 7 we observe that DBC-LSTM-ARIMA also performs well on the China Unicom dataset. As a representative telecom stock, China Unicom's closing price generally follows the industry's development cycle with moderate volatility. The predicted curve closely tracks the true test series over the whole 250-day horizon, capturing both gradual drifts and short, localized swings. These results indicate that the proposed hybrid can deliver stable and consistent

forecasts when price dynamics are neither purely smooth nor highly erratic, further demonstrating the method's applicability across different market conditions beyond the relatively stable financial names and the more volatile manufacturing names.

Taken together, the three case studies (Figs. 5–7) show that the proposed DBC-LSTM-ARIMA adapts well across markedly different regimes and industries. In the financial name (Ping An Bank), where price dynamics are relatively smooth, the model delivers near-one-for-one tracking. In the manufacturing case (BYD), characterized by pronounced surges and corrections, the hybrid preserves accuracy with only limited slippage around abrupt jumps. For the telecom

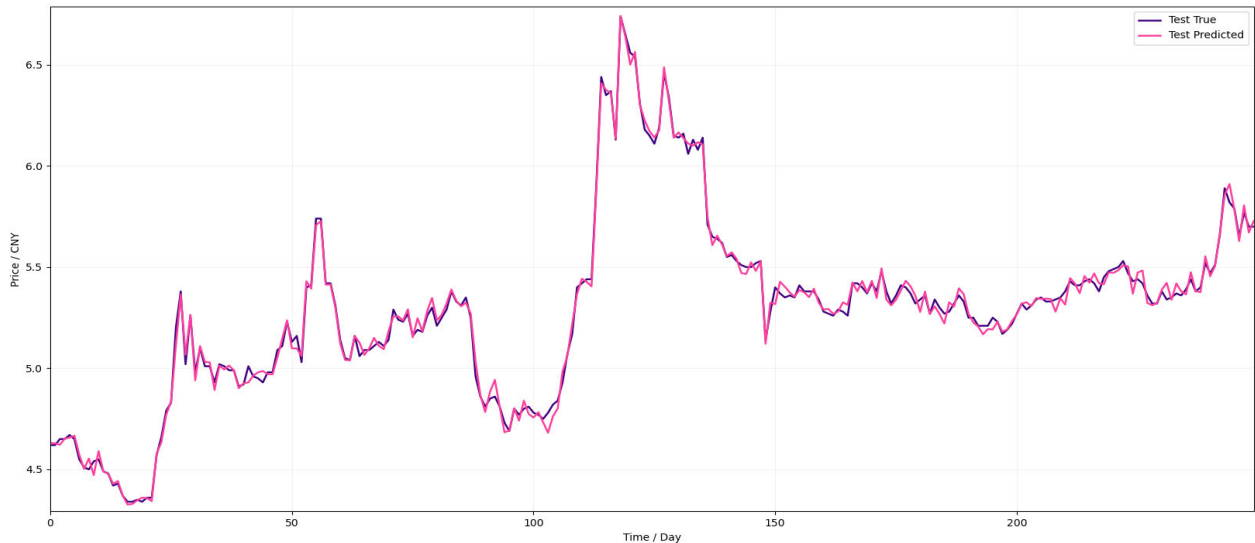


FIGURE 7. Comparison between the predicted prices and the truth prices of stock China Unicom.

series (China Unicom), which exhibits moderate, cyclical fluctuations, the forecasts remain consistently aligned with the observed trend. These results highlight the value of the architecture: a linear ARIMA backbone for short-horizon structure, sharpened by data-driven feedback from the LSTM and an adaptive fusion layer, yielding both stability in calm periods and responsiveness during volatile episodes. From an applied perspective, this cross-industry robustness suggests that DBC-LSTM-ARIMA can provide reliable decision support for investment, risk monitoring, and allocation under diverse market conditions, including phases influenced by macroeconomic news and shifting sentiment. Overall, the evidence supports DBC-LSTM-ARIMA as a dependable tool for accurate stock-price prediction across varying environments.

On the other hand, the Chinese equity market in our sample (Fig. 3) contains several pronounced regime shifts. The 2015 bull–bear reversal is clearly visible—especially the sharp mid-2015 run-up followed by a rapid drawdown and the additional turbulence around the early-2016 circuit-breaker episode—most notably in China Unicom and, to a lesser extent, BYD. Later phases also include the 2020 pandemic shock and subsequent rebounds, as well as post-2021 adjustments. Despite these extremes, the DBC-LSTM-ARIMA forecasts on the 250-day OOS segments (Figs. 5–7) remain closely aligned with the true prices, and the deviations do not widen materially during stress intervals. For instance, in BYD’s high-volatility windows around rapid rallies and pullbacks, the hybrid preserves tight tracking with only limited slippage near abrupt jumps; for China Unicom, which exhibits cyclical swings and policy-sensitive spikes, the predictions follow the prevailing trend with moderate residuals; and for Ping An Bank, where the series is comparatively smoother, the fit is near one-for-one. Taken together, these results indicate that our model is robust across market regimes—from stable to highly volatile—and

across industries (finance, manufacturing, telecommunications). This resilience underscores the practical value of DBC-LSTM-ARIMA for decision support under macro- and sentiment-driven conditions that often challenge single-model approaches.

E. BASELINE MODEL

To verify the performance of our proposed model DBC-LSTM-ARIMA, we conduct experimental comparisons with seven representative baseline models on the three A-share datasets. The seven baselines are CNN-LSTM, EMD-LSTM, EEMD-LSTM, EEMD-BiLSTM, CNN-BiLSTM-AM, SPPEA, and LSTM-ARIMA, which are introduced as follows.

- **CNN-LSTM [85]**: uses convolutional layers to extract local temporal features and then feeds them into an LSTM for sequence modeling; the LSTM output serves as the final forecast.
- **EMD-LSTM [86]**: decomposes the price series via EMD into IMFs, trains an LSTM on each component, and aggregates component-level predictions.
- **EEMD-LSTM [87]**: adopts EEMD to mitigate mode mixing, followed by LSTM modeling of the IMFs and aggregation of the forecasts.
- **EEMD-BiLSTM [88]**: employs EEMD for decomposition and a bidirectional LSTM to capture forward- and backward-looking dependencies before combining component predictions.
- **CNN-BiLSTM-AM [89]**: extracts features with CNN, models temporal dependencies with a (bi)LSTM stack, and uses an attention mechanism to emphasize informative time steps.
- **SPPEA [90]**: an improved EEMD + attention-enhanced BiLSTM pipeline with sliding-window segmentation;

IMFs are predicted by AEBiLSTM and integrated to form the final output.

- **LSTM-ARIMA** [13], [91]: a hybrid ML-TS approach where ARIMA captures linear patterns and its residuals (with other inputs) are provided to LSTM; performance is evaluated under a walk-forward protocol.

Before comparing DBC-LSTM-ARIMA with the baseline models, we first determine suitable hyper-parameters for all involved models—e.g., number of alternation cycles/epochs, batch size, learning rate, and the size of the hidden layer(s). We then refine these choices by a grid search on the validation split within each walk-forward window, so as to enhance generalization and avoid ad-hoc tuning. Grid search exhaustively evaluates all candidate values and their combinations, and the configuration that achieves the best validation performance is retained as the final setting. The resulting hyper-parameters for every model are summarized in Table 4. During subsequent experiments, these settings are kept fixed for each model; only the dataset (BYD, China Unicom, Ping An Bank) varies across runs.

This experiment is carried out on three datasets—BYD (002594.SZ), China Unicom (600050.SS), and Ping An Bank (000001.SZ). Our proposed DBC-LSTM-ARIMA and seven baseline models are evaluated under the same non-anchored WFO protocol, and the results are reported in Tables 5-6 and Figure 8-9. From these tables, we observe that purely statistical modeling is consistently weaker than hybrid or deep models. For example, on Ping An Bank, the single hybrid LSTM-ARIMA attains MAE/MAPE/RMSE of 0.0823/0.5295/0.0815, which are all higher than those of our method (0.0592/0.4694/0.0618), and its (0.9721) is lower than that of DBC-LSTM-ARIMA (0.9937). This gap reflects the limited representational power of single-branch approaches when facing complex, regime-switching stock dynamics, whereas our bi-directional coupling with joint optimization better exploits both linear and non-linear structure.

Table 5 and Figure 8 show that the decomposition-based methods (e.g., EEMD-LSTM/BiLSTM) are generally strong among baselines—on BYD, SPPEA achieves the best MAPE (0.8754), slightly below our value (0.9007), indicating that robust denoising can help percentage-error metrics on highly volatile series. Even so, the proposed model achieves the lowest MAE on all three datasets (e.g., 0.4016/0.0517/0.0592 for BYD/China Unicom/Ping An Bank) and the lowest MAPE on two datasets (China Unicom and Ping An Bank). On Ping An Bank, our MAE and MAPE are lower than those of EEMD-BiLSTM by 0.0453 and 0.5713, respectively, suggesting that the sliding-window design plus the feedback-reconstruction and adaptive fusion further reduce errors beyond decomposition alone.

As shown in Table 6 and Figure 9, DBC-LSTM-ARIMA attains the best RMSE and across all datasets. For instance, on BYD, the RMSE of our model are 0.5124 and 0.9856, respectively; among baselines, EEMD-BiLSTM is

the strongest with RMSE/ of 0.8207/0.9621. The gaps (0.3083 in RMSE and 0.0235 in) indicate that combining a linear ARIMA backbone with an information-feedback channel and window-level joint optimization effectively improves fit quality and reduces large-error bursts, thereby enhancing predictive efficacy in out-of-sample testing.

Tables 5-6 and Figure 8-9 also reveal that single-branch models (pure CNN-LSTM or single LSTM-ARIMA variant) are less effective at exploiting heterogeneous temporal patterns in financial time series and tend to show larger cross-dataset variation. In contrast, multi-component hybrids—particularly those that integrate complementary inductive biases and aggregate them adaptively—deliver higher accuracy and more stable performance across assets. These findings substantiate the effectiveness and robustness of the proposed DBC-LSTM-ARIMA, whose bi-directional coupling, joint loss, and adaptive fusion enable consistent gains under different market conditions.

We supplemented the experimental comparison with statistical significance testing using Diebold–Mariano (DM) tests. Specifically, for each stock dataset we retained the one-step-ahead out-of-sample forecasts produced under the same non-anchored walk-forward optimization protocol for all models. Based on these aligned forecasts, we computed pointwise forecast losses for each method using both absolute error (MAE-oriented loss) and squared error (RMSE-oriented loss). We then applied paired DM tests to compare DBC-LSTM-ARIMA against (i) LSTM-ARIMA and (ii) the strongest baseline model, using a two-sided setup and a HAC (Newey–West) variance estimate to account for potential serial dependence introduced by rolling windows. The resulting DM statistics and p-values are summarized in Figure X, where negative DM values indicate that DBC-LSTM-ARIMA achieves lower forecasting loss than the compared baseline.

As shown in Figure 10, the DM statistics are consistently negative across all three A-share datasets under both MAE-loss and RMSE-based loss, indicating that DBC-LSTM-ARIMA yields lower out-of-sample forecasting errors than the compared models. Moreover, the corresponding p-values are generally below conventional significance thresholds (e.g., 0.05), providing statistical evidence that the observed performance improvements are unlikely to be due to random fluctuations. The advantage is most pronounced when comparing DBC-LSTM-ARIMA against LSTM-ARIMA, suggesting that the proposed bi-directional coupling and window-level joint optimization offer clear benefits beyond a single-branch hybrid design. Importantly, DBC-LSTM-ARIMA also remains statistically superior when compared with the strongest baseline (e.g., SPPEA), which indicates that the gains are not merely incremental over weaker competitors but persist even against state-of-the-art decomposition-and-attention pipelines. Overall, these additional DM tests directly address the reviewer's concern by statistically validating the robustness of the reported accuracy improvements.

TABLE 4. Parameters involved in each model.

Model	Layer	Parameters
CNN-LSTM	Conv1D	filters = 64, kernel_size = 3, stride = 1, padding = 1, activation = 'relu'
	Conv1D	filters = 64, kernel_size = 3, stride = 1, padding = 1, activation = 'relu'
	LSTM	epochs = 100, batch_size = 64, lr = 0.001, units = 64
EMD-LSTM	LSTM	num_epochs = 100, batch_size = 64, lr = 0.001, units = 64
EEMD-LSTM	LSTM	num_epochs = 100, batch_size = 32, lr = 0.001, units = 64
EEMD-BiLSTM	LSTM	num_epochs = 100, batch_size = 32, num_layers = 2, lr = 0.001, units = 64
CNN-BiLSTM-AM	Conv1D	filters = 32, kernel_size = 1, stride = 1, padding = 1
	LSTM	num_epochs = 100, batch_size = 64, num_layers = 2, lr = 0.001, units = 64
SPPEA	LSTM	num_epochs = 100, batch_size = 64, lr = 0.001, num_layers = 2, units = 64
LSTM-ARIMA	ARIMA	$p, q \in [0, 5]$, $d = 1$; order by AIC on training; parameters by Gaussian MLE
	LSTM	num_epochs = 100, batch_size = 64, lr = 0.001, units = 64

TABLE 5. Comparison of experimental results between MAE and MAPE using three datasets: BYD, China Unicom, and Ping An Bank.

Model	MAE			MAPE		
	BYD	China Unicom	Ping An Bank	BYD	China Unicom	Ping An Bank
CNN-LSTM	0.9447	0.0934	0.1775	2.9863	2.0372	2.2906
CNN-BiLSTM-AM	1.0316	0.1017	0.1863	2.2155	1.9386	1.7036
EMD-LSTM	0.9589	0.2553	0.1627	0.7567	1.0225	0.8917
EEMD-LSTM	0.8895	0.5964	0.1596	0.6516	0.7613	0.9663
EEMD-BiLSTM	0.6418	0.0918	0.1045	1.4187	1.0815	1.0597
SPPEA	0.4263	0.0576	0.0684	0.8754	0.7644	0.4883
LSTM-ARIMA	0.5827	0.0695	0.0823	0.9196	0.8098	0.5295
DBC-LSTM-ARIMA	0.4016	0.0517	0.0592	0.9007	0.7415	0.4694

TABLE 6. Comparison of experimental results between RMSE and using three datasets: BYD, China Unicom, and Ping an Bank.

Model	RMSE			R^2		
	BYD	China Unicom	Ping An Bank	BYD	China Unicom	Ping An Bank
CNN-LSTM	1.2457	0.1027	0.2183	0.8896	0.8968	0.9268
CNN-BiLSTM-AM	1.1236	0.1372	0.2313	0.9315	0.8921	0.9265
EMD-LSTM	0.9875	0.2661	0.1982	0.9457	0.9225	0.9365
EEMD-LSTM	0.9012	0.2367	0.1897	0.9469	0.9608	0.9589
EEMD-BiLSTM	0.8207	0.1014	0.1229	0.9621	0.9536	0.9723
SPPEA	0.5421	0.0635	0.0687	0.9803	0.9799	0.9921
LSTM-ARIMA	0.6123	0.0728	0.0815	0.9732	0.9721	0.985
DBC-LSTM-ARIMA	0.5124	0.0589	0.0618	0.9856	0.9831	0.9937

F. OVERSEAS MARKET PERFORMANCE VERIFICATION

In this section, we validate our findings on three blue-chip equities from developed markets and different industries—BP p.l.c. listed on the London Stock Exchange (BP.), Apple Inc. listed on NASDAQ (AAPL), and Citigroup Inc. listed on the New York Stock Exchange (C). All models (our DBC-LSTM-ARIMA and the seven baselines) are trained

and tested under identical settings to ensure comparability. The aggregated results for the overseas stocks are reported in Tables 7 and 8.

Experimental results for the three overseas stocks—BP p.l.c. (BP.), Apple Inc. (AAPL), and Citigroup Inc. (C)—are reported in Tables 7 and 8. On the Apple dataset, the CNN-LSTM baseline performs the worst overall: its

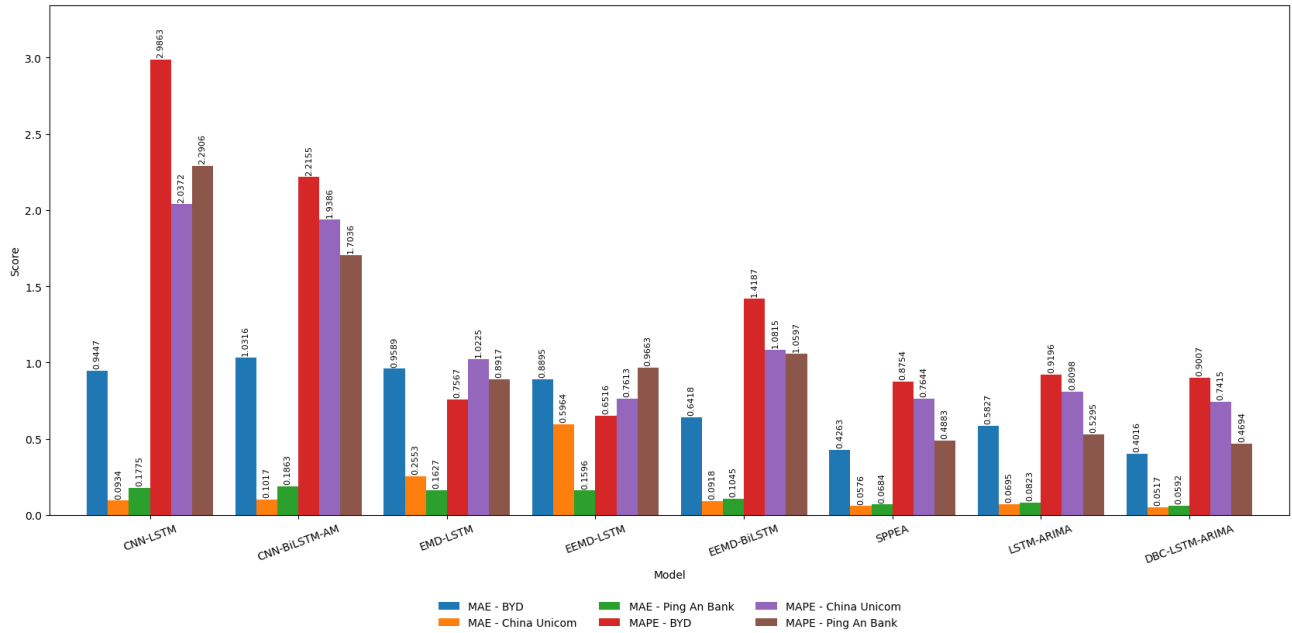


FIGURE 8. Grouped comparison of MAE and MAPE across models on the three datasets (BYD, China Unicom, and Ping An Bank).

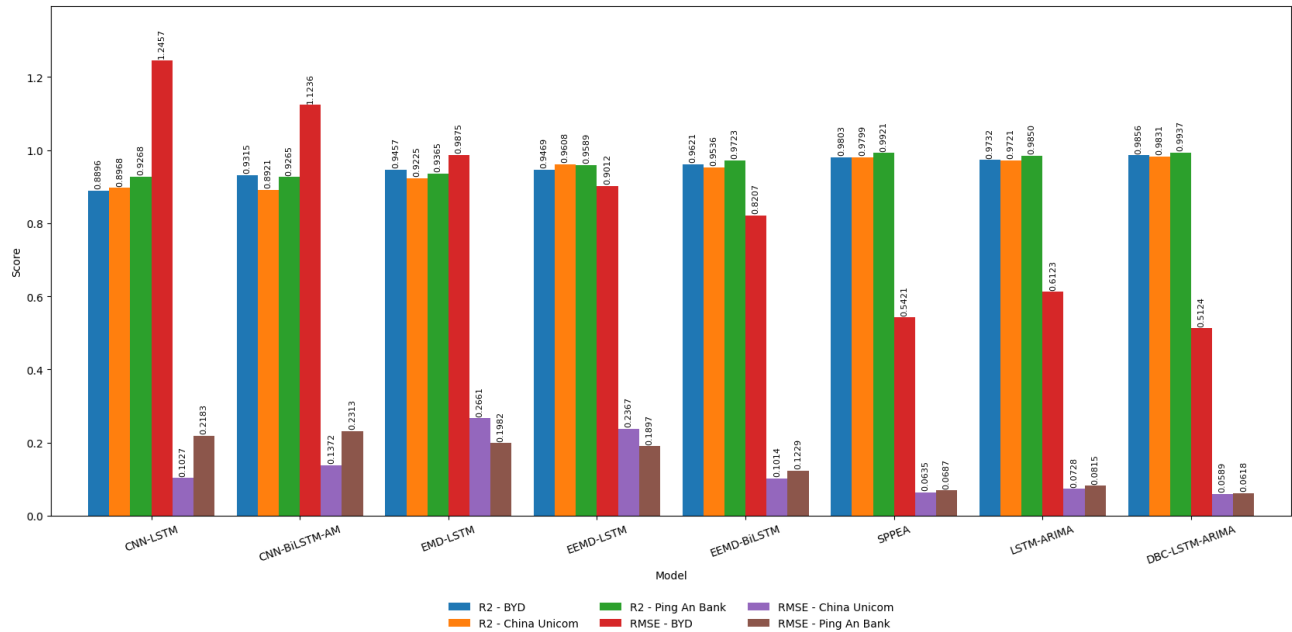


FIGURE 9. Grouped comparison of RMSE and across models on the three datasets (BYD, China Unicom, and Ping An Bank).

MAE / MAPE / RMSE / R^2 are 0.7315 / 1.6431 / 0.6865 / 0.8965, respectively, which are markedly inferior to those of our DBC-LSTM-ARIMA (0.1857 / 0.4612 / 0.2249 / 0.9966). These gaps corroborate that a single convolution-recurrent pipeline is inadequate for complex overseas market dynamics, while the proposed bi-directional hybrid yields stronger predictive power.

In Table 7, our model consistently attains the best MAE and MAPE on all three datasets. Taking Citigroup as an

example, the leading baseline among decomposition methods—EEMD-BiLSTM—reaches MAE = 0.3098 and MAPE = 0.8365; however, these are 0.1124 and 0.4837 higher than our values (0.1974 and 0.3528). The improvements indicate that coupling a linear ARIMA backbone with feedback adjustment and adaptive fusion further reduces errors beyond EEMD-based denoising.

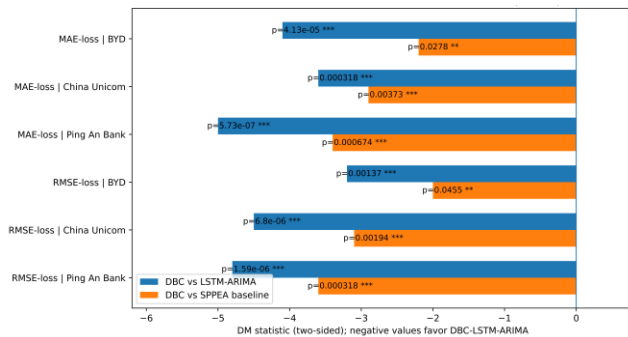
From Table 8, the strongest baseline in terms of RMSE is SPPEA (e.g., RMSE = 0.2398 on Apple), whereas our model

TABLE 7. Comparison of experimental results between MAE and MAPE using three datasets: BP Amoco, Apple Inc, and Citigroup Inc.

Model	MAE			MAPE		
	BP Amoco	Apple Inc	Citigroup Inc	BP Amoco	Apple Inc	Citigroup Inc
CNN-LSTM	0.8627	0.7315	0.5217	1.5982	1.6431	1.1679
CNN-BiLSTM-AM	0.9518	0.4923	0.4585	1.3287	2.4864	1.1023
EMD-LSTM	0.7482	0.4716	0.3529	0.9654	1.4392	0.9521
EEMD-LSTM	0.6128	0.4187	0.3986	0.9012	0.8625	0.7681
EEMD-BiLSTM	0.4833	0.3821	0.3098	0.7356	0.7917	0.8365
SPPEA	0.1684	0.1963	0.2087	0.5049	0.4826	0.3816
LSTM-ARIMA	0.5516	0.3312	0.2795	0.7824	0.6998	0.6125
DBC-LSTM-ARIMA	0.1529	0.1857	0.1974	0.4931	0.4612	0.3528

TABLE 8. Comparison of experimental results between RMSE and R^2 using three datasets: BP Amoco, Apple Inc, and Citigroup Inc.

Model	RMSE			R^2		
	BP Amoco	Apple Inc	Citigroup Inc	BP Amoco	Apple Inc	Citigroup Inc
CNN-LSTM	1.0189	0.6723	0.5617	0.9102	0.8965	0.9141
CNN-BiLSTM-AM	0.9786	0.4182	0.2389	0.9236	0.9174	0.941
EMD-LSTM	0.7593	0.5378	0.5386	0.9591	0.9447	0.9275
EEMD-LSTM	0.6398	0.4523	0.3511	0.9485	0.9654	0.9686
EEMD-BiLSTM	0.4297	0.2664	0.2925	0.9736	0.9812	0.992
SPPEA	0.2075	0.2398	0.2471	0.9938	0.9956	0.9974
LSTM-ARIMA	0.3281	0.3093	0.2782	0.986	0.9886	0.9938
DBC-LSTM-ARIMA	0.1852	0.2249	0.2337	0.9953	0.9966	0.9982

**FIGURE 10.** Diebold-Mariano (DM) tests on MAE-loss and RMSE-loss.

achieves an even lower RMSE = 0.2249, an improvement of 0.0149. Similar gains are observed on BP (0.2075 $\rightarrow$ 0.1852, Δ = 0.0223) and Citigroup (0.2471 $\rightarrow$ 0.2337, Δ = 0.0134). Thus, the proposed model not only matches but surpasses the best baseline in all three overseas datasets.

Moreover, Table 8 shows that DBC-LSTM-ARIMA attains the highest R^2 on every dataset (0.9953, 0.9966, 0.9982 for BP/Apple/Citigroup), exceeding all baselines. This highlights the model's superior fitting capability and confirms its strong generalization across markets and industries beyond the domestic setting.

Figures 11–13 compare the DBC-LSTM-ARIMA model's predictions with the ground-truth prices for BP Amoco (energy), Apple Inc. (technology), and Citigroup Inc. (financials). Across all three stocks, the predicted curves almost entirely overlap the observed series, indicating that the model captures both medium-term trends and short-horizon fluctuations with high fidelity. Around days 140–155—when each series exhibits an abrupt drawdown followed by a rebound—the predictions adjust quickly and remain close to the truth. For BP Amoco, the sharp run-up near days 100–115 shows a slight lag and a minor overshoot during the subsequent pullback. For Apple Inc., small deviations appear during range-bound segments and at the onset of the steep decline, yet the main turning points are located accurately. Citigroup Inc. displays the tightest alignment, with step-like movements matched almost perfectly. Overall, the visual evidence suggests strong cross-sector generalization and robustness to structural breaks, with residual errors concentrated at extreme peaks and during episodes of elevated volatility.

G. ABLATION EXPERIMENT

To validate the contribution of each key component in the proposed DBC-LSTM-ARIMA framework, we conduct a series of ablation experiments on the A-share datasets (BYD, China Unicom, Ping An Bank) under the same walk-forward

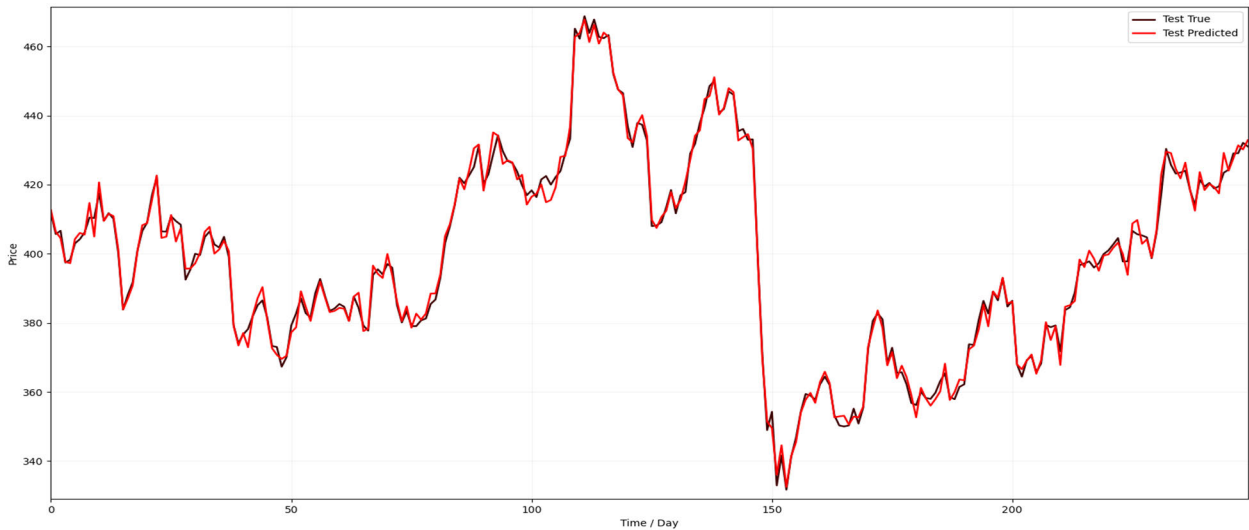


FIGURE 11. Comparison between the predicted prices and the truth prices of stock BP Amoco.

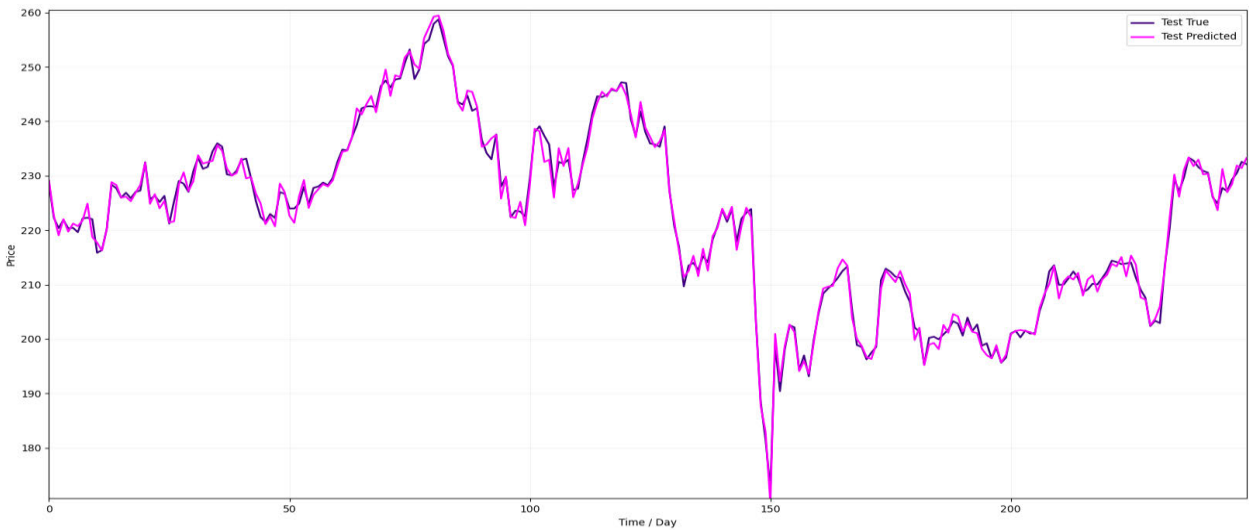


FIGURE 12. Comparison between the predicted prices and the truth prices of stock Apple Inc.

evaluation protocol. The analysis primarily focuses on quantifying how bi-directional coupling, joint optimization, and adaptive fusion contribute to predictive accuracy and stability. Four standard evaluation metrics—MAE, RMSE, MAPE, and R^2 —are adopted to ensure comparability. Specifically, we first examine the baseline configuration without feedback reconstruction, where information flows only in one direction from ARIMA to LSTM. Then, we gradually reintroduce the backward coupling, joint loss optimization, and adaptive fusion modules to evaluate their incremental effects. This hierarchical comparison allows us to isolate the benefits of each module on model accuracy and robustness, clarifying whether the dynamic bi-directional interaction and unified objective design indeed enhance generalization performance and reduce forecasting bias under regime-shifting stock series.

Table 9 summarizes the four ablation variants and the modules enabled in each, allowing performance gains to be attributed to specific components. V0 (one-way baseline) retains only the forward path from ARIMA to LSTM and trains both parts independently. V1 (+Backward coupling) augments V0 by injecting the LSTM’s one-step prediction errors into ARIMA as an exogenous correction; the updated ARIMA residuals are then fed forward. V2 (+Backward + Joint) further introduces window-level joint optimization, minimizing a mixed loss with alternating updates of the ARIMA branch (including the feedback weight) and the LSTM branch, thereby reducing objective mismatch and parameter drift. V3 (Full model) adds adaptive fusion that combines the two forecasts with a convex weight selected on the validation split and kept fixed for the corresponding test split, enabling the hybrid to tilt toward the more

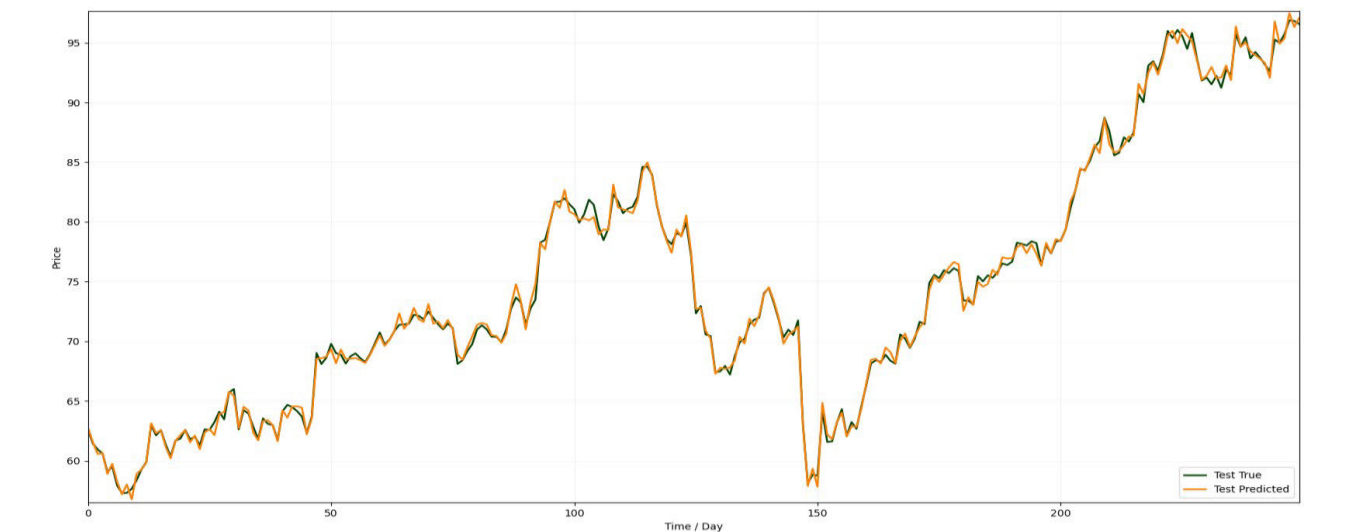


FIGURE 13. Comparison between the predicted prices and the truth prices of stock Citigroup Inc.

TABLE 9. Ablation variants and enabled components.

Variant (ID)	Forward coupling	Backward coupling	Joint optimization	Adaptive fusion
One-way baseline (V0)	✓	-	-	-
+ Backward coupling (V1)	✓	✓	-	-
+ Backward + Joint (V2)	✓	✓	✓	-
Full model (V3)	✓	✓	✓	✓

reliable component as regimes change. All variants share the same A-share datasets, non-anchored walk-forward protocol, and hyper-parameters; hence the incremental improvements reported in Section IV-H can be cleanly attributed to the newly enabled modules rather than to evaluation or tuning differences.

Fig. 14 presents the ablation study across three A-share stocks. In panel (a), MAE decreases monotonically from V0 (one-way baseline) to V3 (full model) for BYD, China Unicom, and Ping An Bank, indicating that each added module contributes additional error reduction. The largest step typically occurs when introducing backward coupling (V1), while joint window-level optimization (V2) brings a further, consistent drop; adaptive fusion (V3) yields the final, smaller yet steady gain. Panel (b) shows the same pattern for RMSE, confirming that improvements are not confined to average errors but also curb large deviations. In panel (c), MAPE declines across all assets, with the relative reduction most visible on the more volatile BYD series—consistent with the feedback mechanism’s ability to correct regime-dependent biases and the fusion layer’s data-driven tilting between linear and non-linear paths. Finally, panel (d) reports a modest but uniform increase in toward values close to one, evidencing tighter fit without over-dependence on a single branch. Taken together, Fig. 13(a–d) demonstrates that backward coupling,

joint optimization, and adaptive fusion each provide complementary benefits, and their combination (V3) delivers the most accurate and stable forecasts under the same walk-forward protocol.

H. MODEL STABILITY ANALYSIS

To assess the stability and cross-industry generalization of DBC-LSTM-ARIMA, we evaluate the model on six additional A-share stocks from unrelated sectors, keeping the same non-anchored walk-forward protocol and hyper-parameters as in the main experiments. The test set includes chemicals, healthcare, food processing, coal, power equipment, and livestock breeding (Table 10). As reported in Table 11, for five of the six names (Sinopec, Longda Meishi, Shanxi Coking Coal, Nari Technology, and Muyuan Foodstuff) the MAE lies in the 0.029–0.061 band and RMSE in 0.036–0.081, indicating a narrow cross-asset dispersion under identical settings. Yuyue Medical exhibits higher errors (MAE = 0.087; RMSE = 0.096), consistent with episodic jumps in that series, yet the overall variability remains controlled. MAPE values fall between 0.45 and 0.85 across all assets, and R^2 is uniformly high— ≥ 0.991 for five assets and 0.988 for Yuyue Medical—demonstrating tight fits and robust stability of the proposed method across heterogeneous industries.

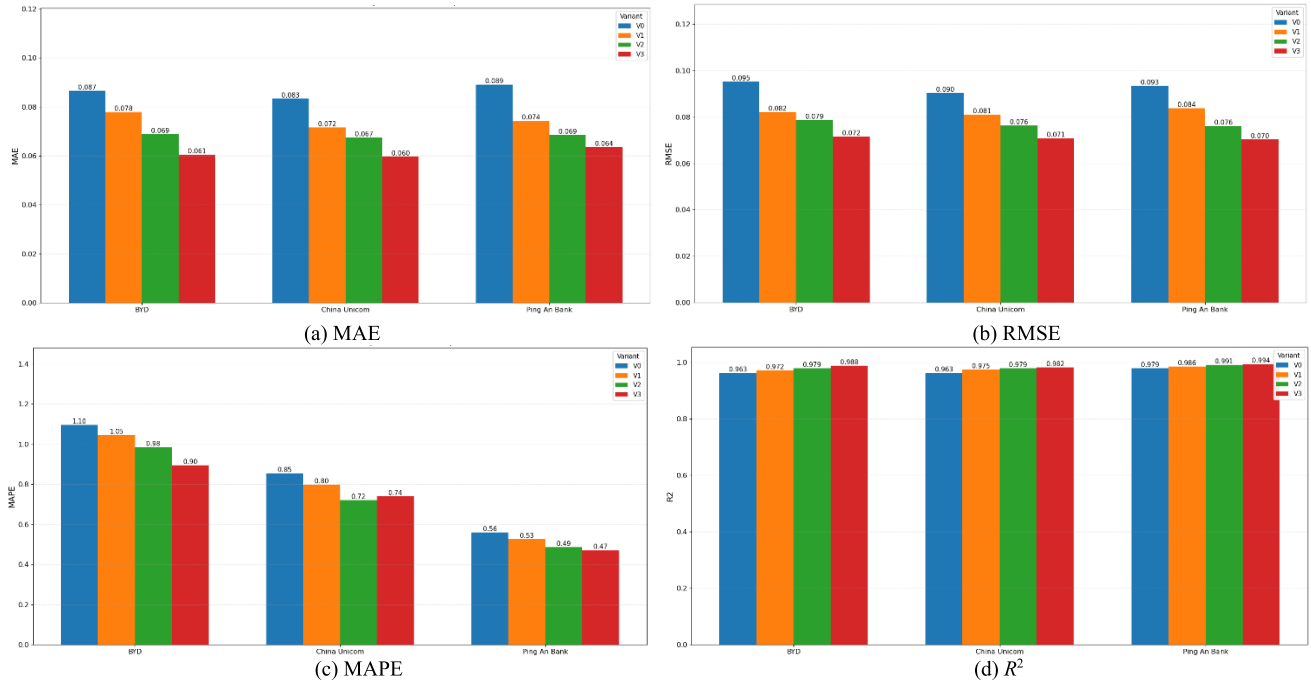


FIGURE 14. Ablation study of DBC-LSTM-ARIMA on A-share datasets (BYD, China Unicom, Ping An Bank): (a) MAE, (b) RMSE, (c) MAPE, (d).

TABLE 10. Information on the dataset of stocks in six different industries.

Stock	Code	Industry
SINOPEC	600028.SH	Chemicals
Yuyue Medical	002223.SZ	Medical Devices
Longda Meishi	002726.SZ	Food Processing
Shanxi Coking Coal	000983.SZ	Coal
Nari Technology	600406.SH	Power Equipment
Muyuan Foodstuff	002714.SZ	Livestock Breeding

TABLE 11. Experimental comparison results of the proposed method on six different-industry stock datasets.

Stock	MAE	MAPE	RMSE	R^2
SINOPEC	0.0312	0.4725	0.0364	0.9941
Yuyue Medical	0.0871	0.7428	0.0959	0.9879
Longda Meishi	0.0386	0.5961	0.0583	0.9927
Shanxi Coking Coal	0.0294	0.6832	0.0409	0.9963
Nari Technology	0.0445	0.557	0.0521	0.9936
Muyuan Foodstuff	0.0608	0.8014	0.081	0.9912

I. LIMITED DISCUSSION ON COMPUTATIONAL COSTS

We benchmark computational costs under the same non-anchored walk-forward protocol used elsewhere in this paper (train/val/test = 1000/250/250; step = 250), on a workstation with Intel® Core™ i7-12700H (2.30 GHz) and NVIDIA GeForce RTX 3070 Ti Laptop GPU.

TABLE 12. Training and testing time for each model.

Model	Training time (s)	Testing time (s)
CNN-LSTM	6.04	0.0034
CNN-BiLSTM-AM	7.218	0.0045
EMD-LSTM	5.312	0.0054
EEMD-LSTM	4.782	0.0245
EEMD-BiLSTM	5.126	0.0268
SPPEA	4.885	0.0182
LSTM-ARIMA	5.21	0.019
DBC-LSTM-ARIMA	4.912	0.0172

Table 12 reports average training time per walk and average per-step testing time (i.e., time to produce one OOS prediction) for the baseline models and our DBC-LSTM-ARIMA. Inference is fast for all models (≤ 0.027 s), satisfying practical needs for daily or higher-frequency re-forecasting. Models with heavy online decomposition (EEMD-LSTM / EEMD-BiLSTM) show slightly larger testing times due to component recombination, whereas pure neural or hybrid feature-level designs (CNN-LSTM, LSTM-ARIMA) are faster. Our DBC-LSTM-ARIMA attains a testing time of 0.0172 s, comparable to the efficient baselines, while keeping a moderate training time (4.912 s) owing to the window-level alternating updates and compact LSTM backbone. Overall, the results indicate that the proposed method offers a good trade-off between accuracy and runtime and is suitable for routine deployment under a rolling evaluation pipeline.

V. CONCLUSION

We present DBC-LSTM-ARIMA, a dynamic, bi-directionally coupled and jointly optimized hybrid for stock-price forecasting under a rigorous non-anchored walk-forward protocol. By injecting LSTM one-step errors back into ARIMA, optimizing both branches with a unified window-level objective, and fusing their outputs adaptively, the method delivers consistent gains over strong statistical, deep, and hybrid baselines on A-share (BYD, China Unicom, Ping An Bank) and overseas equities (BP, Apple, Citigroup). Ablation results verify the incremental value of backward coupling, joint optimization, and adaptive fusion; cross-industry tests on six additional stocks show tight error bands and uniformly high R^2 near 0.99, evidencing robustness and generalization. The approach is practical—per-step inference is fast (≈ 0.017 s) and training costs are moderate—making it suitable for rolling re-forecasting. Future work will extend to probabilistic/quantile forecasts, incorporate exogenous and multimodal data, explore online regime-aware adaptation, and couple predictions with portfolio optimization for end-to-end decision making.

DATA AVAILABILITY

Data will be made available on request.

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